THE NEGATIVE-VISCOSITY EFFECT IN FERROFLUIDS

A Complete First-Principles Reconstruction of Every Equation

VOLUME VIII

The Two-Dimensional Theory:
the Missing 2D Version of the Two Papers

Where Volume VII ended. Volumes I–VII derived every equation of Shliomis & Morozov (1994) and Bacri *et al.* (1995) from first principles in three dimensions. That work is the canonical reference for this volume and is not repeated.

Volume VIII reformulates the entire theory for a genuinely two-dimensional system. Its rule is the opposite of the earlier volumes': where a derivation is unaffected by dimensionality it is *stated* to carry over and the reason given, in one or two lines; where dimensionality enters the mathematics, every step is done afresh in two-dimensional vector calculus, with no step taken by analogy. Several results change in ways that a naive substitution $\mathrm{d}^3r \to \mathrm{d}^2r$ would get *wrong*, and these are flagged explicitly: the Langevin function becomes a ratio of Bessel functions, the rotational friction coefficient acquires a different numerical factor, the Brownian time loses a factor of two, angular velocity ceases to be a vector, and the angle β of (SM-2) disappears from the theory altogether. Chapter numbers resume at 70, page numbers at 240.

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Part XIII

The Two-Dimensional Theory

Chapter 70

What "Two-Dimensional" Means, and the Units

70.1 Two readings, and which one we develop

The phrase "two-dimensional ferrofluid" can mean two different things, and the mathematics differs.

1. **Strictly two-dimensional.** Space is $\mathbb{R}^2$. The carrier is a two-dimensional fluid with an areal mass density and a surface viscosity; the particles are rigid *disks*; the magnetic moments lie in the plane; the magnetic field lies in the plane; the vorticity and the particle spin are out-of-plane pseudoscalars. Every operator, measure and normalisation is the two-dimensional one.

2. **Quasi-two-dimensional (thin film).** A thin layer of ordinary three-dimensional ferrofluid, so thin that the particles cannot move out of plane and their moments are confined to it. Orientational statistics is then genuinely two-dimensional, but hydrodynamics is still three-dimensional: momentum leaks into the surrounding medium.

This volume develops case 1 in full, because that is what "the 2D version of the papers" means, and because case 2 is then obtained from it by one substitution (Sec. 72.7): the strictly-2D results hold in a film provided the particle radius is small compared with the Saffman–Delbrück length. Where the two cases genuinely part company — and there is exactly one such place, the *translational* mobility — it is said so.

70.2 Notation and units

Throughout, d denotes the spatial dimension, and results are quoted for general d wherever a clean general-d form exists, with $d = 3$ recovering the earlier volumes as a check. All two-dimensional quantities carry the same symbols as their three-dimensional counterparts, but their *units* differ, and this must be fixed before anything else.

Symbol	Meaning in 2D	Units (CGS-like)	3D counterpart
ρ	areal mass density of the fluid	$\mathrm{g\,cm^{-2}}$	$\mathrm{g\,cm^{-3}}$
ρ_s	areal mass density of the particle material	$\mathrm{g\,cm^{-2}}$	$\mathrm{g\,cm^{-3}}$
η	surface (2D) shear viscosity	$\mathrm{g\,s^{-1}}$	$\mathrm{g\,cm^{-1}s^{-1}}$
p, σ_{ij}	pressure, stress: force per unit *length*	$\mathrm{dyn\,cm^{-1}}$	$\mathrm{dyn\,cm^{-2}}$
V	"volume" of one particle = its area πa^2	$\mathrm{cm^2}$	$\mathrm{cm^3}$
n	particles per unit area	$\mathrm{cm^{-2}}$	$\mathrm{cm^{-3}}$
$\varphi = nV$	area fraction	—	volume fraction
M	magnetic moment per unit area	$\mathrm{erg\,G^{-1}cm^{-2}}$	G
$m = M_s V$	moment of one particle	$\mathrm{erg\,G^{-1}}$	$\mathrm{erg\,G^{-1}}$
Ω, Θ	**pseudoscalars**, not vectors	$\mathrm{s^{-1}}$	vectors, $\mathrm{s^{-1}}$
ζ_r	rotational friction coefficient	erg s	erg s
kT	thermal energy	erg	erg

Derivation of the units of η. In two dimensions a stress is a force transmitted across a *line*, so $[\sigma] = \mathrm{dyn\,cm^{-1}}$. The Newtonian law $\sigma = \eta\,\partial v/\partial x$ then gives

$$[\eta] = \frac{\mathrm{dyn\,cm^{-1}}}{\mathrm{s^{-1}}} = \mathrm{dyn\,s\,cm^{-1}} = \mathrm{g\,s^{-1}}. \tag{70.2.1}$$

This is the unit of *surface* viscosity, sometimes called the surface poise. It is *not* the poise: dimensionally, $\eta_{2\mathrm{D}} = \eta_{3\mathrm{D}} \times (\text{length})$, and for a film of thickness h the correspondence is

$$\eta_{2\mathrm{D}} = \eta_{3\mathrm{D}}\, h. \tag{70.2.2}$$

Every formula below is dimensionally consistent in these units, and each is checked.

Magnetic units. The Gaussian convention $B = H + 4\pi M$ is a three-dimensional statement: the 4π is the area of the unit sphere. Its two-dimensional counterpart carries the circumference of the unit circle instead (Sec. 72.6),

$$\nabla \cdot \big(\boldsymbol{H} + 2\pi \boldsymbol{M}\big) = 0. \tag{70.2.3}$$

We show in Sec. 72.6 that this constant never enters any viscosity result, so the choice of magnetic convention is immaterial to everything that follows.

70.3 What does *not* change, and why

Three things are worth settling immediately, because they save a great deal of work.

1. **The Boltzmann distribution.** Its derivation (Sec. 9.3) used only reservoir statistics and the definition of temperature. Nothing in it refers to space at all. It carries over verbatim; what changes is the *measure* on the orientation manifold (Ch. 73).

2. **The Einstein relation $D_r = kT/\zeta_r$.** Its derivation (Sec. 11.5) required only that the equilibrium distribution be a zero-flux stationary state of a drift–diffusion equation. That argument is dimension-independent, and the relation is unchanged — although both D_r and ζ_r take different values (Ch. 72, Ch. 74).

3. **The structure of continuum mechanics.** Conservation of mass and the Cauchy momentum equation follow from the divergence theorem, which holds in any dimension. Only the interpretation of the integrals changes (area for volume, line for surface).

Chapter 71

Vector and Tensor Calculus in Two Dimensions

71.1 Context

The three-dimensional theory is built on the cross product, and the cross product does not exist in two dimensions. Everything downstream — vorticity, torque, the antisymmetric stress, the magnetisation equation — must therefore be rebuilt. This chapter does that. It is short but it is the foundation, and every sign in the rest of the volume depends on the conventions fixed here.

71.2 The Levi-Civita symbol in two dimensions

Definition 71.2.1. ε_{ij} $(i, j = 1, 2)$ is totally antisymmetric with $\varepsilon_{12} = +1$; hence $\varepsilon_{21} = -1$ and $\varepsilon_{11} = \varepsilon_{22} = 0$.

The two-index symbol replaces the three-index one, and the identities change accordingly. They are proved by direct enumeration, exactly as in Sec. 1.6.

Result 71.2.1 (The 2D ε–δ identities)**.**

$$\varepsilon_{ij}\varepsilon_{kl} = \delta_{ik}\delta_{jl} - \delta_{il}\delta_{jk}, \tag{71.2.1}$$

$$\varepsilon_{ij}\varepsilon_{kj} = \delta_{ik}, \tag{71.2.2}$$

$$\varepsilon_{ij}\varepsilon_{ij} = 2. \tag{71.2.3}$$

Proof of (71.2.1)**.** Both sides vanish unless $i \neq j$ and $k \neq l$. If $i = k$, $j = l$ (with $i \neq j$) the left side is $(\pm 1)^2 = 1$ and the right side is $1 - 0 = 1$. If $i = l$, $j = k$ the left side is -1 and the right side is $0 - 1 = -1$. All other index sets give zero on both sides. ■

Proof of (71.2.2) **and** (71.2.3)**.** Contract $j = l$ in (71.2.1): $\varepsilon_{ij}\varepsilon_{kj} = \delta_{ik}\delta_{jj} - \delta_{ij}\delta_{jk} = 2\delta_{ik} - \delta_{ik} = \delta_{ik}$, using $\delta_{jj} = 2$ in two dimensions. Contracting once more, $\varepsilon_{ij}\varepsilon_{ij} = \delta_{ii} = 2$. ■

Remark 71.2.1 (A trap). In three dimensions $\varepsilon_{ijk}\varepsilon_{ljk} = 2\delta_{il}$ and $\varepsilon_{ijk}\varepsilon_{ijk} = 6$. Replacing $3 \to 2$ in those numbers gives the wrong answers: the correct two-dimensional statements are δ_{ik} and 2, as just derived. This is the first of several places where a naive reduction fails.

71.3 Products: what a "cross product" becomes

In two dimensions there is no vector perpendicular to the plane, so the cross product of two in-plane vectors cannot be a vector. It is a *pseudoscalar*.

Definition 71.3.1 (Wedge of two vectors)**.**

$$\boldsymbol{a} \wedge \boldsymbol{b} \equiv \varepsilon_{ij} a_i b_j = a_1 b_2 - a_2 b_1. \tag{71.3.1}$$

This is precisely the z component of the three-dimensional cross product of two in-plane vectors, which is the only component such a product can have. It is antisymmetric, $\boldsymbol{a} \wedge \boldsymbol{b} = -\boldsymbol{b} \wedge \boldsymbol{a}$, and $\boldsymbol{a} \wedge \boldsymbol{a} = 0$.

Definition 71.3.2 (Wedge of a pseudoscalar and a vector)**.**

$$\big(S \wedge \boldsymbol{a}\big)_i \equiv S\, \varepsilon_{ji} a_j. \tag{71.3.2}$$

Why this definition and not the other sign. Embed the plane in three dimensions and let S stand for the z component of a vector $S\hat{\boldsymbol{z}}$. Then $(S\hat{\boldsymbol{z}} \times \boldsymbol{a})_i = S\varepsilon_{i3j} a_j$; with $\varepsilon_{132} = -1$ and $\varepsilon_{231} = +1$ this gives $S(-a_2,\, a_1)$. And $S\varepsilon_{ji} a_j$ gives, for $i = 1$, $S\varepsilon_{21} a_2 = -S a_2$, and for $i = 2$, $S\varepsilon_{12} a_1 = +S a_1$. The two agree. ✓ Thus (71.3.2) is the unique definition consistent with the three-dimensional embedding, and it is a counterclockwise rotation by $\pi/2$ scaled by S.

Two consequences used constantly below:

$$\boldsymbol{a} \cdot \big(S \wedge \boldsymbol{a}\big) = S\, \varepsilon_{ji} a_j a_i = 0 \qquad \text{(antisymmetric contracted with symmetric)}, \tag{71.3.3}$$

$$\big(S \wedge \boldsymbol{a}\big) \wedge \boldsymbol{b} = S\, \varepsilon_{ij} \varepsilon_{ki} a_k b_j = -S\, \delta_{jk} a_k b_j = -S\, \big(\boldsymbol{a} \cdot \boldsymbol{b}\big), \tag{71.3.4}$$

the second using $\varepsilon_{ij}\varepsilon_{ki} = -\varepsilon_{ji}\varepsilon_{ki} = -\delta_{jk}$ from (71.2.2).

The triple product, replaced. The identity $\boldsymbol{a} \times (\boldsymbol{a} \times \boldsymbol{b}) = \boldsymbol{a}(\boldsymbol{a} \cdot \boldsymbol{b}) - \boldsymbol{b}\, a^2$ has a two-dimensional counterpart which we shall need in Ch. 75. Let $S = \boldsymbol{a} \wedge \boldsymbol{b}$; then

$$\begin{aligned}
\big(S \wedge \boldsymbol{a}\big)_i &= \varepsilon_{kl} a_k b_l\, \varepsilon_{ji} a_j && && (71.3.5)\\
&= a_k a_j b_l\, \varepsilon_{kl} \varepsilon_{ji} && && (71.3.6)\\
&= a_k a_j b_l \big(\delta_{kj}\delta_{li} - \delta_{ki}\delta_{lj}\big) && \text{by (71.2.1)} && (71.3.7)\\
&= a^2 b_i - a_i \big(\boldsymbol{a} \cdot \boldsymbol{b}\big). && && (71.3.8)
\end{aligned}$$

Note the sign. In three dimensions $\boldsymbol{a} \times (\boldsymbol{a} \times \boldsymbol{b}) = \boldsymbol{a}(\boldsymbol{a} \cdot \boldsymbol{b}) - a^2 \boldsymbol{b}$, which is -1 times (71.3.8). The discrepancy is not an error: the two-dimensional object corresponds to $(\boldsymbol{a} \times \boldsymbol{b}) \times \boldsymbol{a}$, with the factors in the opposite order, because in two dimensions the pseudoscalar must be written on the left of the wedge by (71.3.2). A reader who transcribes the 3D formula mechanically will get the sign of the transverse relaxation term of the magnetisation equation backwards. For that reason we avoid nested wedges entirely below and write the transverse projector out explicitly, $b_i - a_i(\hat{\boldsymbol{a}} \cdot \boldsymbol{b})$, which is unambiguous.

71.4 Differential operators

Operator	Three dimensions	Two dimensions
gradient	$(\nabla f)_i = \partial_i f$	unchanged, $i = 1, 2$
divergence	$\nabla \cdot \boldsymbol{A} = \partial_i A_i$	unchanged, $i = 1, 2$
curl of a vector	vector, $\varepsilon_{ijk}\partial_j A_k$	**pseudoscalar**, $\operatorname{curl} \boldsymbol{A} = \varepsilon_{ij}\partial_i A_j$
curl of a pseudoscalar	—	**vector**, $(\operatorname{curl} S)_i = \varepsilon_{ji}\partial_j S$
Laplacian	$\partial_j\partial_j$	unchanged in form, $j = 1, 2$

The curl is the operator that changes character. In two dimensions it maps vectors to pseudoscalars and pseudoscalars to vectors; it is not an endomorphism, and there is no such thing as "the curl of the curl of a vector" without passing through a scalar. The identity $\nabla \times (\nabla \times \boldsymbol{A}) = \nabla(\nabla \cdot \boldsymbol{A}) - \nabla^2 \boldsymbol{A}$ is replaced by the following, which we derive because it is used in every viscosity calculation.

Result 71.4.1. *For any vector field $\boldsymbol{v}$, with $\Omega \equiv \frac{1}{2}\operatorname{curl} \boldsymbol{v}$,*

$$\varepsilon_{ij}\partial_j \Omega = \tfrac{1}{2}\Big[\partial_i\big(\nabla \cdot \boldsymbol{v}\big) - \nabla^2 v_i\Big] \xrightarrow[\nabla\cdot\boldsymbol{v}=0]{} -\tfrac{1}{2}\nabla^2 v_i. \tag{71.4.1}$$

Derivation.

$$\varepsilon_{ij}\partial_j \Omega = \varepsilon_{ij}\partial_j\Big(\tfrac{1}{2}\varepsilon_{kl}\partial_k v_l\Big) = \tfrac{1}{2}\varepsilon_{ij}\varepsilon_{kl}\,\partial_j\partial_k v_l \tag{71.4.2}$$

$$= \tfrac{1}{2}\big(\delta_{ik}\delta_{jl} - \delta_{il}\delta_{jk}\big)\partial_j\partial_k v_l \qquad \text{by (71.2.1)} \tag{71.4.3}$$

$$= \tfrac{1}{2}\big(\partial_j\partial_i v_j - \partial_j\partial_j v_i\big) = \tfrac{1}{2}\Big[\partial_i\big(\nabla \cdot \boldsymbol{v}\big) - \nabla^2 v_i\Big]. \quad \blacksquare \tag{71.4.4}$$

Numerical check. Take plane Poiseuille flow $\boldsymbol{v} = (v(y), 0)$. Then $\Omega = \frac{1}{2}(\partial_1 v_2 - \partial_2 v_1) = -\frac{1}{2}v'(y)$, and $\varepsilon_{ij}\partial_j\Omega$ has components $\big(\varepsilon_{12}\partial_2\Omega,\ \varepsilon_{21}\partial_1\Omega\big) = \big(-\frac{1}{2}v'',\, 0\big)$, while $-\frac{1}{2}\nabla^2\boldsymbol{v} = \big(-\frac{1}{2}v'',\, 0\big)$. ✓

71.5 Local decomposition of a flow, and the pseudoscalar vorticity

The Taylor expansion and the symmetric/antisymmetric split of $\partial_j v_i$ are algebraic and carry over unchanged. What changes is the identification of the antisymmetric part. In two dimensions an antisymmetric 2×2 matrix has exactly *one* independent component, so

$$a_{ij} = \tfrac{1}{2}\big(\partial_j v_i - \partial_i v_j\big) = \Omega\,\varepsilon_{ji}, \tag{71.5.1}$$

and contracting with ε_{ji} and using (71.2.3),

$$\boxed{\Omega = \tfrac{1}{2}\varepsilon_{ij}\partial_i v_j = \tfrac{1}{2}\operatorname{curl} \boldsymbol{v}} \tag{71.5.2}$$

with $a_{ij}\delta_j = (\Omega \wedge \boldsymbol{\delta})_i$ by (71.3.2). The formula "$\Omega = \frac{1}{2}\operatorname{curl}\boldsymbol{v}$" therefore survives *verbatim*; only the type of the object changes, from vector to pseudoscalar.

Remark 71.5.1 (Why this matters more than it looks). The rotation group in two dimensions is $SO(2)$, which is one-dimensional and abelian. Consequences that run through the whole volume:

- There is no angle β between $\boldsymbol{H}$ and Ω: the field lies in the plane, the vorticity is normal to it, and the two are *always* perpendicular. The factor $\sin^2\beta$ of (SM-2) is identically 1 (Sec. 77.5).
- Rotations commute, so $\Theta \wedge (\Omega \wedge \boldsymbol{a}) = \Omega \wedge (\Theta \wedge \boldsymbol{a})$ and no ordering ambiguities arise.
- A single scalar Ω replaces three components, so the spin field carries one degree of freedom instead of three.

71.6 Isotropic averages, measures and transforms

These are collected here because they are used repeatedly and because each is a place where dimensionality enters directly.

Result 71.6.1 (Isotropic angular averages)**.** *For a unit vector $\boldsymbol{n}$ uniformly distributed on the unit circle,*

$$\langle n_i n_j \rangle = \tfrac{1}{2}\delta_{ij}, \qquad \langle n_i n_j n_k n_l \rangle = \tfrac{1}{8}\big(\delta_{ij}\delta_{kl} + \delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk}\big). \tag{71.6.1}$$

Derivation. By isotropy $\langle n_i n_j \rangle = A\delta_{ij}$; taking the trace, $\langle n_i n_i \rangle = 1 = A\delta_{ii} = 2A$, so $A = \tfrac{1}{2}$. For the fourth-rank average, isotropy forces the three-δ combination with a single coefficient C; contracting $i = j$ and $k = l$ gives $1 = C(\delta_{ii}\delta_{kk} + 2\delta_{ik}\delta_{ik}) = C(4 + 4) = 8C$, so $C = \tfrac{1}{8}$. In general d the coefficients are $1/d$ and $1/[d(d+2)]$, giving $\tfrac{1}{3}$ and $\tfrac{1}{15}$ in three dimensions. ✓

Object	Three dimensions	Two dimensions
volume measure	$\mathrm{d}^3 r$	$\mathrm{d}^2 r$
orientation measure	$\mathrm{d}\Omega_e = \sin\theta\,\mathrm{d}\theta\,\mathrm{d}\phi$	$\mathrm{d}\phi$
total measure	4π	2π
Fourier	$\int \dfrac{\mathrm{d}^3 k}{(2\pi)^3}$	$\int \dfrac{\mathrm{d}^2 k}{(2\pi)^2}$
delta function	$\delta^3(\boldsymbol{r})$, units cm^{-3}	$\delta^2(\boldsymbol{r})$, units cm^{-2}
Laplace Green fn.	$\nabla^2 G = \delta$, $G = -\dfrac{1}{4\pi r}$	$G = \dfrac{1}{2\pi}\ln r$
isotropic $\langle n_i n_j \rangle$	$\tfrac{1}{3}\delta_{ij}$	$\tfrac{1}{2}\delta_{ij}$

Derivation of the 2D Green's function. Seek a radial $G(r)$ with $\nabla^2 G = 0$ for $r > 0$. In polar coordinates $\nabla^2 G = \frac{1}{r}\frac{\mathrm{d}}{\mathrm{d}r}\big(r\frac{\mathrm{d}G}{\mathrm{d}r}\big) = 0$ gives $r\,\mathrm{d}G/\mathrm{d}r = \text{const} = c$, hence $G = c\ln r$. Fix c by integrating $\nabla^2 G = \delta^2$ over a disk of radius r and using the divergence theorem: $1 = \oint \nabla G \cdot \boldsymbol{n}\,\mathrm{d}l = (c/r)(2\pi r) = 2\pi c$, so $c = 1/2\pi$. ■

The logarithm, rather than an inverse power, is the single most consequential mathematical difference between two and three dimensions, and it is responsible for the Stokes paradox (Sec. 72.4) and for the long-time-tail problem (Sec. 83.4).

Chapter 72

Two-Dimensional Hydrodynamics and Magnetostatics

72.1 What carries over

Conservation of mass, the Cauchy momentum equation and the Newtonian constitutive law are derived from the divergence theorem and from linearity, neither of which refers to dimension. They carry over with "volume" read as "area" and "surface" as "bounding curve":

$$\nabla \cdot \boldsymbol{v} = 0, \tag{72.1.1}$$

$$\rho \frac{\mathrm{d}v_i}{\mathrm{d}t} = f_i + \partial_j \sigma_{ij}, \qquad \sigma_{ij} = -p\,\delta_{ij} + 2\eta e_{ij} + \sigma^a_{ij}, \tag{72.1.2}$$

with $e_{ij} = \frac{1}{2}(\partial_j v_i + \partial_i v_j)$ and $i, j = 1, 2$. Substituting the symmetric part gives the two-dimensional Navier–Stokes equation, identical in form to Eq. (5.4.5).

One subtlety in the constitutive law. In d dimensions the trace-free rate of strain is $e_{ij} - \frac{1}{d}\delta_{ij} e_{kk}$, so the split into shear and bulk parts uses $\frac{1}{2}$ in 2D where 3D uses $\frac{1}{3}$. For an incompressible fluid $e_{kk} = 0$ and the distinction is empty, which is why (72.1.2) looks unchanged. If compressibility were ever restored, this coefficient would have to be corrected.

72.2 The Reynolds number

Unchanged in form, $\mathrm{Re} = \rho U \ell / \eta$, but now with areal density and surface viscosity. Dimensional check:

$$\big[\mathrm{Re}\big] = \frac{(\mathrm{g\,cm^{-2}})(\mathrm{cm\,s^{-1}})(\mathrm{cm})}{\mathrm{g\,s^{-1}}} = 1. \quad \checkmark \tag{72.2.1}$$

For colloidal particles it is again astronomically small, and the Stokes equations $0 = -\nabla p + \eta \nabla^2 \boldsymbol{v}$, $\nabla \cdot \boldsymbol{v} = 0$ apply.

72.3 Rotational friction of a disk: the central hydrodynamic result

Result 72.3.1. *A rigid disk of radius a rotating with angular velocity Θ in an unbounded two-dimensional Stokes fluid experiences the retarding torque*

$$\boxed{T_{visc} = -\zeta_r \Theta, \qquad \zeta_r = 4\pi\eta a^2 = 4\eta V} \tag{72.3.1}$$

with $V = \pi a^2$ the particle area.

Step 1: the flow. Try the purely azimuthal field

$$v_\phi(r) = \frac{\Theta a^2}{r}, \qquad v_r = 0. \tag{72.3.2}$$

It satisfies the no-slip condition $v_\phi(a) = \Theta a$ and decays at infinity. It is incompressible ($v_r = 0$ and v_ϕ independent of ϕ). It is also *irrotational* for $r > 0$: $\operatorname{curl} \boldsymbol{v} = \frac{1}{r}\frac{\mathrm{d}}{\mathrm{d}r}(r v_\phi) = \frac{1}{r}\frac{\mathrm{d}}{\mathrm{d}r}(\Theta a^2) = 0$. A field that is both divergence-free and curl-free satisfies $\nabla^2 \boldsymbol{v} = 0$ by (71.4.1), so the Stokes equation holds with constant pressure.

Remark 72.3.1 (This is the point-vortex field). Equation (72.3.2) is the two-dimensional rotlet, and it is the same field as an ideal point vortex. Its decay is $1/r$, slower than the $1/r^2$ of the three-dimensional rotating sphere — yet, as Step 3 shows, the torque integral still converges. Slow decay is not by itself fatal; what matters is the decay of the *stress*.

Step 2: the stress. For a purely azimuthal, axisymmetric flow the only nonzero shear component is

$$\sigma_{r\phi} = \eta\, r \frac{\partial}{\partial r}\left(\frac{v_\phi}{r}\right) = \eta\, r \frac{\partial}{\partial r}\left(\frac{\Theta a^2}{r^2}\right) = \eta\, r\left(-\frac{2\Theta a^2}{r^3}\right) = -\frac{2\eta\Theta a^2}{r^2}, \tag{72.3.3}$$

so at the surface $\sigma_{r\phi}(a) = -2\eta\Theta$. Note it is independent of a.

Step 3: the torque.

$$T = \oint a\, \sigma_{r\phi}\big|_{r=a}\, \mathrm{d}l = a\big(-2\eta\Theta\big)\big(2\pi a\big) = -4\pi\eta a^2\Theta, \tag{72.3.4}$$

which is (72.3.1). ■

Independent check by energy dissipation. The dissipation rate per unit area is $2\eta e_{ij}e_{ij}$, and for this flow $e_{r\phi} = \frac{1}{2} r\,\partial_r(v_\phi/r) = -\Theta a^2/r^2$, so $e_{ij}e_{ij} = 2e_{r\phi}^2 = 2\Theta^2 a^4/r^4$. Hence

$$\dot{W} = \int_a^\infty 4\eta \frac{\Theta^2 a^4}{r^4}\, 2\pi r\, \mathrm{d}r = 8\pi\eta\Theta^2 a^4 \int_a^\infty \frac{\mathrm{d}r}{r^3} = 8\pi\eta\Theta^2 a^4 \cdot \frac{1}{2a^2} = 4\pi\eta a^2\Theta^2. \tag{72.3.5}$$

Since $\dot{W} = |T|\,\Theta$, this gives $|T| = 4\pi\eta a^2\Theta$. ✓ The integral converges at infinity, which is the precise statement that the rotational problem is well posed.

The general-d pattern. Writing ζ_r through the particle "volume" V_d,

$$\zeta_r = 2d\,\eta V_d: \qquad d=3:\ 6\eta V = 8\pi\eta a^3; \qquad d=2:\ 4\eta V = 4\pi\eta a^2. \tag{72.3.6}$$

Consequently, per unit area,

$$\boxed{n\zeta_r = 4\eta\varphi} \qquad (\text{3D: } 6\eta\varphi). \tag{72.3.7}$$

Every factor of $6\eta\varphi$ in Volumes I–VII becomes $4\eta\varphi$ here.

72.4 The Stokes paradox, and why it does not damage the theory

The *translational* problem has no counterpart. Seeking a two-dimensional Stokes flow past a disk that is uniform at infinity and satisfies no-slip leads to a logarithmically growing solution and no finite drag coefficient: the Stokes paradox. The three-dimensional formula $F = -6\pi\eta a\boldsymbol{u}$ therefore has *no* 2D analogue — it is one of the few equations of the earlier volumes that becomes invalid rather than merely modified.

Why nothing is lost. Inspect where translational drag was used in Volumes I–VII: nowhere. The theory uses only (i) the rotational friction ζ_r, (ii) the rotational diffusion coefficient D_r derived from it, and (iii) the assumption that particle and fluid translate together, which is a statement about scale separation, not about drag. The magnetoviscous theory is therefore *better* conditioned in two dimensions than one might fear: the one divergent transport coefficient never appears.

Why the difference exists. The two problems differ in the decay of the disturbance. Rotation produces a $1/r$ velocity and a $1/r^2$ stress, giving a convergent torque; translation would require a disturbance decaying like the 2D Green's function, i.e. logarithmically, and momentum cannot be shed to infinity at that rate.

72.5 The inertial spin time τ_s

The derivation of Ch. 7 — exponential relaxation of the slip with time constant I_p/ζ_r — is a statement about a linear ODE and carries over unchanged. Only the two ingredients change.

Moment of inertia of a disk. With areal density ρ_s,

$$I_p = \int_0^a r^2\,\rho_s\,2\pi r\,\mathrm{d}r = 2\pi\rho_s\frac{a^4}{4} = \frac{\pi\rho_s a^4}{2} = \tfrac{1}{2}m_p a^2, \tag{72.5.1}$$

with $m_p = \rho_s\pi a^2$. (In 3D it was $\frac{2}{5}m_p a^2$: the factor is $2/(d+2)$, giving $\frac{2}{5}$ and $\frac{1}{2}$.)

Result.

$$\tau_s = \frac{I_p}{\zeta_r} = \frac{\pi\rho_s a^4/2}{4\pi\eta a^2} = \frac{\rho_s a^2}{8\eta} = \boxed{\frac{\rho_s d^2}{32\,\eta}} \qquad (\text{3D: } \rho_s d^2/60\eta). \tag{72.5.2}$$

Units: $(\mathrm{g\,cm^{-2}})(\mathrm{cm^2})/(\mathrm{g\,s^{-1}}) = \mathrm{s}$. ✓

Is inertia still negligible? The physically meaningful statement is the ratio to the Brownian time, which by (74.2.3) below is

$$\frac{\tau_s}{\tau_B} = \frac{\rho_s a^2/8\eta}{4\pi\eta a^2/kT} = \frac{\rho_s\, kT}{32\pi\,\eta^2}, \tag{72.5.3}$$

which is *independent of particle size* — a purely 2D feature, since in three dimensions the same ratio carries a factor d^2. Dimensional check: $(\mathrm{g\,cm^{-2}})(\mathrm{g\,cm^2 s^{-2}})/(\mathrm{g^2 s^{-2}}) = 1$. ✓ For a film made from a fluid with $\eta_{3\mathrm{D}} = 10^{-2}$ P and thickness 10 nm one has $\eta \approx 10^{-8}\ \mathrm{g\,s^{-1}}$ and $\rho_s \approx 6\times10^{-6}\ \mathrm{g\,cm^{-2}}$, giving $\tau_s/\tau_B \sim 2\times10^{-9}$: inertia is negligible, as in three dimensions, but for a reason that no longer involves the particle size.

72.6 Two-dimensional magnetostatics

The constraints. The quasi-static, non-conducting reduction of Maxwell's equations is unchanged in logic. What changes is the type of the curl and the constant in the constitutive relation:

$$\operatorname{curl}\boldsymbol{H} = \varepsilon_{ij}\partial_i H_j = 0, \qquad \nabla\cdot\big(\boldsymbol{H} + 2\pi\boldsymbol{M}\big) = 0. \tag{72.6.1}$$

The first is a single scalar equation (three in 3D); it still implies $\boldsymbol{H} = -\nabla\psi$, by the same path-independence argument. The constant 2π in the second is the circumference of the unit circle, replacing the 4π that is the area of the unit sphere; it comes from the Gauss law $\nabla\cdot\boldsymbol{E} = S_d\rho$ with $S_2 = 2\pi$, $S_3 = 4\pi$.

The dipole field. With $\boldsymbol{H} = -\nabla\psi$, (72.6.1) gives $\nabla^2\psi = 2\pi\nabla\cdot\boldsymbol{M}$. For a point dipole $\boldsymbol{m}$ at the origin, $\boldsymbol{M} = \boldsymbol{m}\,\delta^2(\boldsymbol{r})$, and convolving with the Green's function $\ln r/2\pi$ of Sec. 71.6,

$$\psi(\boldsymbol{r}) = -\boldsymbol{m}\cdot\nabla\ln r = -\frac{\boldsymbol{m}\cdot\boldsymbol{r}}{r^2}, \qquad H \sim \frac{m}{r^2} \qquad (\text{3D: } H\sim m/r^3). \tag{72.6.2}$$

Consequence for the dilute assumption. The dipolar interaction energy of two particles a distance r apart scales as m^2/r^d in d dimensions, while the number of neighbours in a shell $\mathrm{d}r$ scales as $nr^{d-1}\mathrm{d}r$. The mean interaction is therefore $\propto \int \mathrm{d}r/r$ in *both* dimensions: dipolar coupling is marginal — logarithmically sensitive to the cutoffs — in 2D exactly as in 3D. The dilute assumption is thus no worse in two dimensions than in three, which is a nontrivial and reassuring result: one might have guessed that the lower-dimensional system is more strongly coupled, and it is not.

Demagnetising fields. The uniform-field assumption is preserved by the same argument as in Sec. 22.6: for a strip magnetised along its long axis, $\nabla\cdot\boldsymbol{M} = 0$ in the interior and the end poles are far away, so the internal field equals the applied one and the constant 2π of (72.6.1) never appears in any result below. Magnetised *across* the strip, the demagnetising factor would be 2π per unit M (the 2D analogue of 4π for a slab).

72.7 The quasi-two-dimensional case: Saffman–Delbrück

A physically realisable "2D ferrofluid" is a thin film of surface viscosity η embedded in a bulk fluid of viscosity η_b. The bulk provides a momentum sink and thereby cures the Stokes paradox, at the price of introducing a length

$$\ell_{\mathrm{SD}} = \frac{\eta}{2\eta_b} \tag{72.7.1}$$

beyond which the outer fluid dominates.

The important fact for this volume is that the *rotational* friction is insensitive to this: for $a \ll \ell_{\mathrm{SD}}$ the rotational drag of a disk inclusion in a membrane is $4\pi\eta a^2$, exactly the strictly-2D value (72.3.1), with corrections of relative order a/ℓ_{SD}. That the membrane literature's rotational result coincides with the derivation of Sec. 72.3 is an independent confirmation of (72.3.1).

Quantity	Strictly 2D	Film, $a \ll \ell_{\mathrm{SD}}$	Film, $a \gg \ell_{\mathrm{SD}}$
rotational drag	$4\pi\eta a^2$	$4\pi\eta a^2$	$\to$ bulk-dominated
translational drag	undefined (paradox)	$\sim 4\pi\eta/[\ln(\ell_{\mathrm{SD}}/a) - \gamma]$	$\to 16\eta_b a/3$
theory of this volume	valid	valid	requires 3D treatment

Chapter 73

Orientational Statistics on the Circle

73.1 The measure and the partition function

The orientation of a rigid in-plane dipole is a single angle, $\boldsymbol{e} = (\cos\phi, \sin\phi)$, $\phi \in [0, 2\pi)$. The orientation manifold is the circle S^1, and its measure is simply

$$\mathrm{d}\mu = \mathrm{d}\phi, \qquad \int \mathrm{d}\mu = 2\pi \qquad (\text{3D: } \sin\theta\,\mathrm{d}\theta\,\mathrm{d}\phi, \text{ total } 4\pi). \tag{73.1.1}$$

There is no Jacobian factor: the $\sin\theta$ of three dimensions came from the shrinking of circles of latitude, and a circle has no latitudes. This single change propagates into every statistical result of the theory.

With $U = -mH\cos\phi$ and $\xi = mH/kT$, the Boltzmann law of Sec. 9.3 gives

$$W_0(\phi) = \frac{e^{\xi\cos\phi}}{Z_2(\xi)}, \qquad Z_2(\xi) = \int_0^{2\pi} e^{\xi\cos\phi}\,\mathrm{d}\phi = 2\pi I_0(\xi), \tag{73.1.2}$$

where I_0 is the modified Bessel function of the first kind of order zero. This is its *definition*: $I_\nu(\xi) = \frac{1}{\pi}\int_0^\pi e^{\xi\cos\phi}\cos\nu\phi\,\mathrm{d}\phi$ for integer ν, so that $\int_0^{2\pi} e^{\xi\cos\phi}\mathrm{d}\phi = 2\pi I_0(\xi)$. There is no elementary closed form — the elementary $\sinh\xi/\xi$ of three dimensions was a consequence of the $\sin\theta$ measure, which allowed the substitution $u = \cos\theta$ and turned the integral into an exponential. In two dimensions no such substitution exists, and the answer is intrinsically a Bessel function.

73.2 The two-dimensional Langevin function

Result 73.2.1.

$$\boxed{\big\langle \cos\phi \big\rangle = L_2(\xi) \equiv \frac{I_1(\xi)}{I_0(\xi)}} \tag{73.2.1}$$

Derivation. By the same parameter-differentiation trick as in Sec. 10.2,

$$\big\langle \cos\phi \big\rangle = \frac{1}{Z_2}\frac{\mathrm{d}Z_2}{\mathrm{d}\xi} = \frac{\mathrm{d}}{\mathrm{d}\xi}\ln\big(2\pi I_0\big) = \frac{I_0'(\xi)}{I_0(\xi)} = \frac{I_1(\xi)}{I_0(\xi)}, \tag{73.2.2}$$

the last step using the standard recurrence $I_0' = I_1$, which follows from differentiating the series $I_\nu(\xi) = \sum_k (\xi/2)^{2k+\nu}/[k!(k+\nu)!]$ term by term. ■

Weak fields. From the series, $I_0 = 1 + \frac{\xi^2}{4} + \frac{\xi^4}{64}$ and $I_1 = \frac{\xi}{2} + \frac{\xi^3}{16} + \cdots$, so

$$L_2(\xi) = \frac{\frac{\xi}{2}\big(1 + \frac{\xi^2}{8}\big)}{1 + \frac{\xi^2}{4}} = \frac{\xi}{2}\left(1 + \frac{\xi^2}{8}\right)\left(1 - \frac{\xi^2}{4} + \cdots\right) = \boxed{\frac{\xi}{2} - \frac{\xi^3}{16} + O(\xi^5)} \tag{73.2.3}$$

against $L_3 = \xi/3 - \xi^3/45$. The leading coefficient is $1/d$: it is $\langle\cos^2\rangle$ over the sphere S^{d-1}, which is $1/2$ on a circle and $1/3$ on a sphere. *Numerically verified:* at $\xi = 0.1$ the exact ratio $I_1/I_0 = 0.0499376$ against the series value 0.0499375.

Strong fields. Using the asymptotic expansion $I_\nu(\xi) \sim \dfrac{e^\xi}{\sqrt{2\pi\xi}}\Big(1 - \dfrac{4\nu^2 - 1}{8\xi}\Big)$,

$$L_2(\xi) = \frac{1 - \frac{3}{8\xi}}{1 + \frac{1}{8\xi}} + \cdots = 1 - \frac{1}{2\xi} + O\big(\xi^{-2}\big) \qquad \text{(3D: } 1 - 1/\xi\text{)}. \tag{73.2.4}$$

Numerically verified: at $\xi = 50$, $I_1/I_0 = 0.98995$ against $1 - 1/(2\xi) = 0.99$. The approach to saturation is *twice as fast* in two dimensions, because there are fewer transverse directions into which thermal agitation can knock the moment.

73.3 The second moment

Result 73.3.1.

$$\big\langle \cos^2\phi \big\rangle = 1 - \frac{L_2(\xi)}{\xi}, \qquad \big\langle e_i e_j \big\rangle = \frac{L_2}{\xi}\delta_{ij} + \left(1 - \frac{2L_2}{\xi}\right) n_i n_j. \tag{73.3.1}$$

Derivation. Write $\cos^2\phi = \frac{1}{2}(1 + \cos 2\phi)$ and use $\langle\cos 2\phi\rangle = I_2/I_0$ (from the integral representation with $\nu = 2$). The recurrence $I_{\nu-1}(\xi) - I_{\nu+1}(\xi) = \frac{2\nu}{\xi} I_\nu(\xi)$ with $\nu = 1$ gives $I_0 - I_2 = \frac{2}{\xi} I_1$, hence

$$\frac{I_2}{I_0} = 1 - \frac{2}{\xi}\frac{I_1}{I_0} = 1 - \frac{2L_2}{\xi}, \tag{73.3.2}$$

and therefore $\langle\cos^2\phi\rangle = \frac{1}{2}\big(1 + 1 - 2L_2/\xi\big) = 1 - L_2/\xi$. The tensor form then follows exactly as in Sec. 61.4: writing $\langle e_i e_j\rangle = A\delta_{ij} + B n_i n_j$, the trace gives $2A + B = 1$ (*note* $\delta_{ii} = 2$, *not 3*) and the longitudinal projection gives $A + B = 1 - L_2/\xi$; subtracting, $A = L_2/\xi$ and $B = 1 - 2L_2/\xi$. ■

Numerically verified by direct quadrature: at $\xi = 2$, $\langle\cos^2\phi\rangle = 0.6511127$ from the integral and $1 - L_2/\xi = 0.6511127$ from the formula; likewise at $\xi = 0.3$ and $\xi = 8$ to seven digits.

General-d form. The pattern is

$$\big\langle e_i e_j \big\rangle = \frac{L_d}{\xi}\delta_{ij} + \left(1 - \frac{d\,L_d}{\xi}\right) n_i n_j, \tag{73.3.3}$$

with $\langle(\boldsymbol{e}\cdot\boldsymbol{n})^2\rangle = 1 - (d-1)L_d/\xi$. Both the coefficient of $n_i n_j$ and the relation to $\langle\cos^2\rangle$ change with d; substituting $3 \to 2$ in only one of them — the commonest error — violates the trace condition.

Limits. As $\xi \to 0$, $L_2/\xi \to \frac{1}{2}$ so $A \to \frac{1}{2}$, $B \to 0$, recovering the isotropic average $\frac{1}{2}\delta_{ij}$ of (71.6.1). ✓ As $\xi \to \infty$, $A \to 0$ and $B \to 1$: full alignment. ✓

73.4 Magnetisation and susceptibility

With $\boldsymbol{M} = nm\langle \boldsymbol{e} \rangle$,

$$\boxed{\boldsymbol{M}_0 = n\, m\, L_2(\xi)\, \hat{\boldsymbol{h}} = \varphi M_s\, \frac{I_1(\xi)}{I_0(\xi)}\, \hat{\boldsymbol{h}}} \tag{73.4.1}$$

and in weak fields, using (73.2.3),

$$\boxed{\boldsymbol{M}_0 = \chi \boldsymbol{H}, \qquad \chi = \frac{nm^2}{2\,kT}} \qquad \text{(3D: } nm^2/3kT\text{)}. \tag{73.4.2}$$

The Curie law is unchanged; the numerical coefficient is $1/d$.

Chapter 74

Rotational Brownian Motion in Two Dimensions

74.1 The Smoluchowski equation on the circle

Orientation space is one-dimensional, so the surface gradient, divergence and Laplacian of Sec. 11.2 all collapse to ordinary derivatives with respect to ϕ. No metric factors survive:

$$\nabla_s \to \frac{\partial}{\partial\phi}, \qquad \nabla_{s\cdot} \to \frac{\partial}{\partial\phi}, \qquad \nabla_s^2 \to \frac{\partial^2}{\partial\phi^2}. \tag{74.1.1}$$

This is a genuine simplification: the awkward $\frac{1}{\sin\theta}\partial_\theta(\sin\theta\,\cdot)$ of three dimensions has no counterpart, and Green's identity becomes ordinary integration by parts over a period.

The torque on a particle whose moment is at angle ϕ in a field along $\hat{\boldsymbol{x}}$ is the pseudoscalar

$$T = \boldsymbol{m}\wedge\boldsymbol{H} = \varepsilon_{ij}m_iH_j = -mH\sin\phi = -\frac{\partial U}{\partial\phi}, \qquad U = -mH\cos\phi, \tag{74.1.2}$$

so the identity "torque $= -\partial U/\partial\phi$" holds exactly as in three dimensions. Conservation of probability and the inertialess torque balance then give

$$\boxed{\frac{\partial W}{\partial t} = -\frac{\partial}{\partial\phi}\left[W\left(\Omega - \frac{1}{\zeta_r}\frac{\partial U}{\partial\phi}\right) - D_r\frac{\partial W}{\partial\phi}\right]} \tag{74.1.3}$$

The Einstein relation. Setting $\Omega = 0$ and demanding that $W_0 \propto e^{-U/kT}$ carry zero flux gives, exactly as in Sec. 11.5,

$$D_r = \frac{kT}{\zeta_r} = \frac{kT}{4\pi\eta a^2} \qquad \text{(unchanged in form; new value).} \tag{74.1.4}$$

74.2 The relaxation time: where a factor of two is lost

Result 74.2.1.

$$\nabla_s^2\boldsymbol{e} = \frac{\partial^2}{\partial\phi^2}\big(\cos\phi,\sin\phi\big) = -\big(\cos\phi,\sin\phi\big) = -\boldsymbol{e}. \tag{74.2.1}$$

The eigenvalue is -1, not -2. In general the vector harmonics on S^{d-1} have eigenvalue $-(d-1)$: -2 on the sphere, -1 on the circle. **This is the single most easily missed change in the whole volume.**

Consequence. Repeating Sec. 12.2, in zero field

$$\frac{\mathrm{d}\langle \boldsymbol{e}\rangle}{\mathrm{d}t} = D_r \int W\, \nabla_s^2 \boldsymbol{e}\, \mathrm{d}\phi = -D_r \langle \boldsymbol{e}\rangle, \tag{74.2.2}$$

so that

$$\boxed{\tau_B = \frac{1}{D_r} = \frac{\zeta_r}{kT} = \frac{4\pi\eta a^2}{kT} = \frac{4\eta V}{kT}} \tag{74.2.3}$$

against $\tau_B = 1/(2D_r) = 3\eta V/kT$ in three dimensions. In general

$$\tau_B = \frac{1}{(d-1)D_r} = \frac{2d\,\eta V_d}{(d-1)kT}: \qquad d=3:\ \frac{6\eta V}{2kT} = \frac{3\eta V}{kT}; \qquad d=2:\ \frac{4\eta V}{kT}. \quad \checkmark \tag{74.2.4}$$

Units: $(\mathrm{g\,s^{-1}})(\mathrm{cm^2})/(\mathrm{g\,cm^2 s^{-2}}) = \mathrm{s}$. ✓

Remark 74.2.1 (The trap, stated explicitly). A naive reduction would take $\tau_B = 3\eta V/kT$ and simply reinterpret V as an area, giving $3\eta V/kT$. That is wrong by the factor $4/3$, for two independent reasons that partly cancel: the friction coefficient changes from $6\eta V$ to $4\eta V$ (a factor $2/3$), and the eigenvalue changes from 2 to 1 (a factor 2). Only by deriving both does one obtain $4\eta V/kT$.

Chapter 75

The Magnetisation Equations in Two Dimensions

75.1 The exact first-moment equation

Multiply (74.1.3) by e_i and integrate over the period. Two ingredients are needed, both special to two dimensions:

$$\frac{\partial e_i}{\partial \phi} = \varepsilon_{ji} e_j \qquad \text{(a counterclockwise rotation by } \pi/2\text{)}, \tag{75.1.1}$$

verified componentwise — $\varepsilon_{j1} e_j = -e_2 = -\sin\phi = \partial_\phi \cos\phi$ and $\varepsilon_{j2} e_j = e_1 = \cos\phi = \partial_\phi \sin\phi$ ✓— and the deterministic angular velocity

$$\dot\phi_{\mathrm{det}} = \Omega + \frac{T}{\zeta_r} = \Omega + D_r\,(\boldsymbol{e} \wedge \boldsymbol{\xi}), \qquad \boldsymbol{\xi} \equiv \frac{m\boldsymbol{H}}{kT}, \tag{75.1.2}$$

where $T/\zeta_r = (m/\zeta_r)\varepsilon_{ij} e_i H_j = D_r \varepsilon_{ij} e_i \xi_j$ by the Einstein relation.

Integrating by parts (no boundary terms: the circle has no boundary),

$$\frac{\mathrm{d}\langle e_i\rangle}{\mathrm{d}t} = \Big\langle \frac{\partial e_i}{\partial \phi}\,\dot\phi_{\mathrm{det}} \Big\rangle + D_r \Big\langle \frac{\partial^2 e_i}{\partial \phi^2} \Big\rangle \tag{75.1.3}$$

$$= \Omega\,\varepsilon_{ji}\langle e_j\rangle + D_r \big\langle \varepsilon_{ji} e_j\, \varepsilon_{ab} e_a \xi_b \big\rangle - D_r \langle e_i\rangle. \tag{75.1.4}$$

The middle term is evaluated with (71.2.1):

$$\varepsilon_{ji}\varepsilon_{ab}\,\xi_b \big\langle e_j e_a \big\rangle = \big(\delta_{ja}\delta_{ib} - \delta_{jb}\delta_{ia}\big)\xi_b \big\langle e_j e_a \big\rangle \tag{75.1.5}$$

$$= \xi_i \big\langle e_j e_j \big\rangle - \xi_j \big\langle e_j e_i \big\rangle = \xi_i - \big\langle e_i e_j \big\rangle \xi_j, \tag{75.1.6}$$

using $\langle e_j e_j \rangle = 1$. Hence

$$\boxed{\frac{\mathrm{d}\langle e_i\rangle}{\mathrm{d}t} = \Omega\,\varepsilon_{ji}\langle e_j\rangle + D_r\Big[\xi_i - \big\langle e_i e_j\big\rangle \xi_j\Big] - D_r\,\langle e_i\rangle} \tag{2D-1}$$

This is *exact*. Comparing with the three-dimensional Eq. (62.3.1): the flow term becomes a wedge, the magnetic term keeps the coefficient D_r, and the relaxation coefficient is D_r instead of $2D_r$. In general d the last coefficient is $(d-1)D_r$, and the magnetic coefficient is D_r in every dimension.

75.2 The phenomenological (isotropic) closure

Setting $\langle e_i e_j\rangle = \frac{1}{2}\delta_{ij}$ — the 2D isotropic value — in (2D-1):

$$D_r\Big[\xi_i - \tfrac{1}{2}\xi_i\Big] = \tfrac{1}{2}D_r\xi_i = D_r\big\langle e_i\big\rangle_{\mathrm{eq}}, \tag{75.2.1}$$

since $\langle \boldsymbol{e}\rangle_{\mathrm{eq}} = L_2\hat{\boldsymbol{h}} \simeq \boldsymbol{\xi}/2$ in weak fields by (73.2.3). With $D_r = 1/\tau_B$ this gives, after multiplying by nm,

$$\boxed{\frac{\mathrm{d}\boldsymbol{M}}{\mathrm{d}t} = \Theta\wedge\boldsymbol{M} - \frac{1}{\tau_B}\big(\boldsymbol{M}-\boldsymbol{M}_0\big)} \tag{2D-2}$$

— formally identical to (SM-7). The replacement of Ω by the full particle spin Θ is justified exactly as in Sec. 13.3, using

$$\boxed{\Theta = \Omega + \frac{1}{4\eta\varphi}\big(\boldsymbol{M}\wedge\boldsymbol{H}\big)} \tag{2D-3}$$

the 2D counterpart of (SM-11), with $4\eta\varphi$ from (72.3.7). Note that $\boldsymbol{M}\wedge\boldsymbol{H}$ is a pseudoscalar, so (2D-3) is a *scalar* equation: the three components of (SM-11) collapse to one.

Where the closure is exact. As in three dimensions, only for $\xi\to 0$ — the isotropic value $\frac{1}{2}\delta_{ij}$ is the leading term of (73.3.1). The correction is $O(\xi^2)$ here as there.

75.3 The effective-field closure

Assume, as in Ch. 61, that the distribution retains its equilibrium form in an effective field $\boldsymbol{\xi}_e = \xi_e\boldsymbol{n}$. Then $\langle\boldsymbol{e}\rangle = L_e\boldsymbol{n}$ with $L_e = L_2(\xi_e) = I_1(\xi_e)/I_0(\xi_e)$, and $\langle e_ie_j\rangle$ is given by (73.3.1). Substituting into (2D-1):

$$\xi_i - \big\langle e_ie_j\big\rangle\xi_j = \xi_i - \frac{L_e}{\xi_e}\xi_i - \left(1-\frac{2L_e}{\xi_e}\right)n_i\big(\boldsymbol{n}\cdot\boldsymbol{\xi}\big). \tag{75.3.1}$$

Now add and subtract to isolate the longitudinal relaxation, exactly as in Sec. 62.4. Writing $-\frac{1}{\tau_B}\frac{L_e}{\xi_e}(\boldsymbol{\xi}_e-\boldsymbol{\xi}) = -\frac{1}{\tau_B}L_e\boldsymbol{n} + \frac{1}{\tau_B}\frac{L_e}{\xi_e}\boldsymbol{\xi}$ and subtracting it from the right-hand side leaves

$$D_r\boldsymbol{\xi}\left(1-\frac{2L_e}{\xi_e}\right) - D_r\left(1-\frac{2L_e}{\xi_e}\right)\boldsymbol{n}\big(\boldsymbol{n}\cdot\boldsymbol{\xi}\big) = -D_r\left(1-\frac{2L_e}{\xi_e}\right)\Big[\boldsymbol{n}\big(\boldsymbol{n}\cdot\boldsymbol{\xi}\big)-\boldsymbol{\xi}\Big], \tag{75.3.2}$$

so that

$$\boxed{\frac{\mathrm{d}}{\mathrm{d}t}\big(L_e\boldsymbol{n}\big) = \Omega\wedge\big(L_e\boldsymbol{n}\big) - \frac{1}{\tau_B}\frac{L_e}{\xi_e}\big(\boldsymbol{\xi}_e-\boldsymbol{\xi}\big) - \frac{1}{\tau_B}\left(1-\frac{2L_e}{\xi_e}\right)\Big[\boldsymbol{n}\big(\boldsymbol{n}\cdot\boldsymbol{\xi}\big)-\boldsymbol{\xi}\Big]} \tag{2D-4}$$

the two-dimensional counterpart of (B-6). Two coefficients changed and both were derived: the prefactor is $1/\tau_B = D_r$ where three dimensions had $1/2\tau_B = D_r$ (the same D_r, differently expressed), and the anisotropy factor is $1-2L_e/\xi_e$ where three dimensions had $1-3L_e/\xi_e$ — i.e. $1-dL_d/\xi_e$, from the trace condition of (73.3.3).

On writing the last term as a double wedge. We deliberately keep the transverse projector explicit. Sec. 71.3 showed that the naive transcription $\boldsymbol{\xi}_e \times (\boldsymbol{\xi}_e \times \boldsymbol{\xi}) \to$ nested 2D wedges reverses the sign, because the pseudoscalar must stand on the left. The form printed in (2D-4) is unambiguous, and in two dimensions it is also simpler, since $\boldsymbol{n}(\boldsymbol{n}\cdot\boldsymbol{\xi}) - \boldsymbol{\xi} = -\boldsymbol{\xi}_\perp$ where $\boldsymbol{\xi}_\perp$ is the single transverse component — there is only one transverse direction.

Zero-approximation and first-order equations. With $\Omega = 0$ and $\boldsymbol{\xi}_e \parallel \boldsymbol{\xi}$ the transverse term vanishes identically and (2D-4) reduces, by the chain rule exactly as in Sec. 63.4, to

$$\tau_B \frac{\mathrm{d}\xi_e}{\mathrm{d}t} = -\left(\frac{\mathrm{d}\ln L_e}{\mathrm{d}\xi_e}\right)^{-1}\left(1 - \frac{\xi_0}{\xi_e}\cos\omega t\right), \qquad L_e = \frac{I_1(\xi_e)}{I_0(\xi_e)}, \tag{2D-5}$$

identical in form to (B-7) — the derivation used only the chain rule and the collinearity, neither of which knows about dimension. Only the function L_e has changed. Its weak- and strong-field behaviour, needed for the same discussion as Sec. 63.5, is

$$\left(\frac{\mathrm{d}\ln L_2}{\mathrm{d}\xi}\right)^{-1} \to \xi \quad (\xi \ll 1), \qquad \to 2\xi^2 \quad (\xi \gg 1), \tag{75.3.3}$$

the second following from $L_2 \simeq 1 - 1/2\xi$, whence $\mathrm{d}L_2/\mathrm{d}\xi \simeq 1/2\xi^2$ and $\mathrm{d}\ln L_2/\mathrm{d}\xi \simeq 1/2\xi^2$. The stiffening at large fields is therefore *twice as strong* as in three dimensions, where the same quantity tends to ξ^2.

Repeating Sec. 64.4 line for line with the 2D coefficients — the only changes being $2L_e \to$ in place of $3L_e$ and D_r in place of $\frac{1}{2}\tau_B^{-1}$ — gives the first-order equation

$$\tau_B \frac{\mathrm{d}F}{\mathrm{d}t} = 1 - \left(\frac{1}{L_e} - \frac{1}{\xi_e}\right)\xi_0 F\cos\omega t, \tag{2D-6}$$

the counterpart of (B-8) with the factor $\frac{1}{2}$ replaced by 1 — again the general-d factor $1/(d-1)$.

Chapter 76

From Torque to Viscosity in Two Dimensions

76.1 A body couple is an antisymmetric stress

Result 76.1.1. *A body-couple density Λ (a pseudoscalar, torque per unit area) is equivalent to the antisymmetric stress and force density*

$$\sigma^a_{ij} = \tfrac{1}{2}\varepsilon_{ij}\Lambda, \qquad f_i = \partial_j \sigma^a_{ij} = \tfrac{1}{2}\varepsilon_{ij}\partial_j\Lambda. \tag{76.1.1}$$

Derivation. The torque exerted on a region by surface tractions is

$$\oint \varepsilon_{ij} x_i \sigma_{jk} n_k \,\mathrm{d}l = \int \partial_k\big(\varepsilon_{ij} x_i \sigma_{jk}\big)\mathrm{d}A = \int \varepsilon_{ij}\sigma_{ji}\,\mathrm{d}A + \int \varepsilon_{ij} x_i \big(\partial_k \sigma_{jk}\big)\mathrm{d}A, \tag{76.1.2}$$

the second integral being the torque already accounted for by the linear-momentum balance. The first is the internal torque density $\varepsilon_{ij}\sigma_{ji}$, and with negligible intrinsic spin angular momentum it must cancel the imposed couple: $\varepsilon_{ij}\sigma^a_{ji} + \Lambda = 0$. Putting $\sigma^a_{ji} = c\,\varepsilon_{ji}\Lambda$ and using (71.2.3),

$$\varepsilon_{ij}\sigma^a_{ji} = c\,\varepsilon_{ij}\varepsilon_{ji}\Lambda = -c\,\varepsilon_{ij}\varepsilon_{ij}\Lambda = -2c\Lambda, \tag{76.1.3}$$

so $c = \frac{1}{2}$, giving (76.1.1). ■

Cross-check against the 3D embedding. Identify Λ with Λ_z of a three-dimensional couple along $\hat{\boldsymbol{z}}$. Then $f_i = \frac{1}{2}(\nabla\times\boldsymbol{\Lambda})_i = \frac{1}{2}\varepsilon_{ij3}\partial_j\Lambda$, and $\varepsilon_{123} = +1 = \varepsilon_{12}$, $\varepsilon_{213} = -1 = \varepsilon_{21}$, so $\varepsilon_{ij3} = \varepsilon_{ij}$ and the two expressions coincide. ✓

76.2 The master formula

Step 1: torque balance. Inertialess particles, per unit area:

$$\boxed{4\eta\varphi\big(\Theta - \Omega\big) = \boldsymbol{M}\wedge\boldsymbol{H}} \tag{2D-7}$$

so the couple delivered to the fluid is $\Lambda = 4\eta\varphi(\Theta - \Omega)$.

Step 2: define the slip. With $\Omega - \Theta = g\,\Omega$ and g a constant scalar, $\Lambda = -4\eta\varphi g\,\Omega$.

Step 3: the extra force. By (76.1.1) and then (71.4.1),

$$f_i = \tfrac{1}{2}\varepsilon_{ij}\partial_j\big(-4\eta\varphi g\Omega\big) = -2\eta\varphi g\,\varepsilon_{ij}\partial_j\Omega = -2\eta\varphi g\,\big(-\tfrac{1}{2}\nabla^2 v_i\big) = +\eta\varphi g\,\nabla^2 v_i. \tag{76.2.1}$$

Step 4: combine with the viscous term.

$$\eta\nabla^2\boldsymbol{v} + \eta\varphi g\nabla^2\boldsymbol{v} = \eta\big(1+\varphi g\big)\nabla^2\boldsymbol{v}, \tag{76.2.2}$$

whence

$$\boxed{\Delta\eta = \eta\varphi\, g = \eta\varphi\,\frac{\Omega-\Theta}{\Omega}} \tag{2D-8}$$

against $\frac{3}{2}\eta\varphi\, g$ in three dimensions. The general-d statement, obtained by the same four steps with $n\zeta_r = 2d\eta\varphi$, is

$$\Delta\eta = \frac{d}{2}\,\eta\varphi\, g: \qquad d=3:\ \tfrac{3}{2}\eta\varphi g; \qquad d=2:\ \eta\varphi g. \quad \checkmark \tag{76.2.3}$$

Sign check. In a saturating stationary field the particles are locked, $\Theta = 0$, $g = 1$, and $\Delta\eta = +\eta\varphi > 0$: the viscosity *increases*, as it must. ✓ Had the sign of (76.1.1) been taken from the 3D formula without the embedding check of Sec. 76.1, this test would have failed.

Saturation value.

$$\Delta\eta(\infty, 0) = \eta\varphi \qquad (\text{3D: } \tfrac{3}{2}\eta\varphi). \tag{76.2.4}$$

The maximum magnetoviscous increment is *smaller* in two dimensions, by the factor $d/2$. Physically: the same number of blocked rotors is embedded in a fluid whose momentum transport is organised differently, and the geometric factor relating the couple to the shear stress is $d/2$.

Chapter 77

The Stationary Field

77.1 The calculation

Geometry: $\boldsymbol{H} = H\hat{\boldsymbol{x}}$ in plane, Ω out of plane, hence automatically perpendicular. Write $\boldsymbol{M} = M_0\hat{\boldsymbol{x}} + m_1\hat{\boldsymbol{y}}$ with $m_1 = O(\Omega\tau_B)$.

Step 1: torque and spin. $\boldsymbol{M} \wedge \boldsymbol{H} = \varepsilon_{ij}M_iH_j = M_xH_y - M_yH_x = -m_1H$, so by (2D-3), $\Theta = \Omega - m_1H/(4\eta\varphi)$.

Step 2: the stationary equation. With $(\Theta \wedge \boldsymbol{M})_i = \Theta\varepsilon_{ji}M_j$, i.e. $\Theta(-m_1,\, M_0)$, the components of (2D-2) with $\mathrm{d}\boldsymbol{M}/\mathrm{d}t = 0$ are

$$\hat{\boldsymbol{y}}: \qquad 0 = \Theta M_0 - \frac{m_1}{\tau_B} \quad\Longrightarrow\quad m_1 = \tau_B\Theta M_0, \tag{77.1.1}$$

$$\hat{\boldsymbol{x}}: \qquad 0 = -\Theta m_1 - \frac{M_x - M_0}{\tau_B} \quad : \text{second order, discarded.} \tag{77.1.2}$$

Step 3: solve. Substituting Θ from Step 1 into (77.1.1),

$$m_1 = \tau_B M_0\left(\Omega - \frac{m_1H}{4\eta\varphi}\right) \quad\Longrightarrow\quad m_1 = \frac{\tau_B M_0\Omega}{1+\zeta}, \qquad \zeta \equiv \frac{\tau_B M_0 H}{4\eta\varphi}. \tag{77.1.3}$$

Step 4: evaluate ζ. This is where the new τ_B and the new ζ_r enter. With $\tau_B = 4\eta V/kT$, $M_0 = nmL_2$, $\varphi = nV$:

$$\zeta = \frac{1}{4\eta(nV)}\cdot\frac{4\eta V}{kT}\cdot n\,m\,L_2\,H = \frac{mH}{kT}L_2(\xi) = \boxed{\xi\,L_2(\xi)} \tag{77.1.4}$$

against $\zeta = \frac{1}{2}\xi L_3(\xi)$ in three dimensions. Every factor of η, n and V cancels, exactly as in Sec. 15.5 — the cancellation is a structural feature, not a coincidence of dimension. The general form is $\zeta = \xi L_d/(d-1)$.

Step 5: the slip and the viscosity. $\Omega - \Theta = m_1H/(4\eta\varphi)$, so

$$g = \frac{\Omega - \Theta}{\Omega} = \frac{H}{4\eta\varphi\,\Omega}\cdot\frac{\tau_B M_0\Omega}{1+\zeta} = \frac{\zeta}{1+\zeta}, \tag{77.1.5}$$

the vorticity cancelling — so the 2D fluid is Newtonian too. ✓ Hence

$$\boxed{\Delta\eta = \eta\varphi\,\frac{\xi L_2(\xi)}{1+\xi L_2(\xi)}, \qquad L_2(\xi) = \frac{I_1(\xi)}{I_0(\xi)}} \tag{2D-9}$$

77.2 No hyperbolic closed form

In three dimensions the corresponding expression collapsed to $(\xi - \tanh\xi)/(\xi + \tanh\xi)$ because $L_3 = (\xi - \tanh\xi)/(\xi\tanh\xi)$ is an elementary function. *No such collapse occurs in two dimensions.* There is no elementary expression for I_1/I_0, and (2D-9) is as closed as the answer gets.

This is a real qualitative difference and it traces directly to the measure: the $\sin\theta$ of the sphere permits the substitution $u = \cos\theta$ and an elementary integral; the flat measure $\mathrm{d}\phi$ of the circle does not.

77.3 Limits and checks

Weak field. $L_2 \simeq \xi/2$, so $\zeta \simeq \xi^2/2 \ll 1$ and

$$\boxed{\Delta\eta \simeq \tfrac{1}{2}\,\eta\varphi\,\xi^2} \qquad (\text{3D: } \tfrac{1}{4}\eta\varphi\xi^2). \tag{77.3.1}$$

General d: $\Delta\eta = \eta\varphi\xi^2/[2(d-1)]$, which gives $\frac{1}{4}$ and $\frac{1}{2}$. ✓ The weak-field effect is *twice as large* in two dimensions at equal ξ and φ.

Strong field. $L_2 \to 1$, $\zeta \to \xi \to \infty$, $g \to 1$, and $\Delta\eta \to \eta\varphi$, matching (76.2.4). ✓

Zero field. $\xi = 0 \Rightarrow \Delta\eta = 0$. ✓

Positivity. $L_2 > 0$ for $\xi > 0$, so $\zeta > 0$ and $\Delta\eta > 0$ for all fields: a stationary field always thickens a 2D ferrofluid, as thermodynamics requires (a permanent magnet does no work). ✓

77.4 The disappearance of β

Equation (SM-2) carries a factor $\sin^2\beta$, with β the angle between $\boldsymbol{H}$ and Ω, and vanishes for $\beta = 0$. **In two dimensions this factor is absent, and no orientation of the field can switch the effect off.** The reason is structural: Ω is not a vector in the plane but a pseudoscalar normal to it, so $\boldsymbol{H}$ and Ω are perpendicular by construction and $\beta \equiv \pi/2$.

Consequences.

- The three-dimensional experimental requirement that the capillary be aligned with the coil axis has no 2D counterpart — any in-plane field direction works equally well.
- The 2D theory has *one fewer parameter*: the response is a function of $(\xi, \omega\tau_B)$ only, with no angular dependence at all.

- A 2D ferrofluid always operates at the maximum of the 3D angular factor. This partly offsets the smaller prefactor $d/2$ of (76.2.3): at $\beta = \pi/2$ the 3D saturation is $\frac{3}{2}\eta\varphi$ against $\eta\varphi$ in 2D, but averaged over field orientations the 3D value would be $\frac{3}{2}\eta\varphi\langle\sin^2\beta\rangle = \eta\varphi$ — the same number.

Chapter 78

The Alternating Field at Low Amplitude

78.1 What carries over untouched

The entire chain of Volumes IV and V — the decomposition of a linearly polarised field into two counter-rotating circular ones, the rotating ansatz, the phase lag, the rotational Doppler shift $\omega \mp \Omega$, the linearisation in $\Omega\tau_B$, the double-angle collapse, and the two-scale averaging — involves *only* (i) rotations in the plane of the field and (ii) scalar response functions. In two dimensions the field still rotates in the (only) plane, and the counter-rotating decomposition

$$\boldsymbol{H}_\pm = \big(H_0 \cos\omega t,\ \pm H_0 \sin\omega t\big), \qquad \boldsymbol{H} = \tfrac{1}{2}\big(\boldsymbol{H}_+ + \boldsymbol{H}_-\big) \tag{2D-10}$$

is meaningful and exact. Every step of Chs. 30–41 therefore carries over *verbatim*, with three substitutions and no others:

$$6\eta\varphi \to 4\eta\varphi, \qquad \chi = \frac{nm^2}{3kT} \to \frac{nm^2}{2kT}, \qquad \tau_B = \frac{3\eta V}{kT} \to \frac{4\eta V}{kT}. \tag{78.1.1}$$

In particular the central intermediate result is unchanged in form:

$$M_y = \Omega\tau_B\, \chi H_0 \cos^2\alpha\, \cos\big(\omega t - 2\alpha\big), \qquad \tan\alpha = \omega\tau_B. \tag{2D-11}$$

The lag is still *twice* the single-component lag, for the same reason (the response is a difference of two oppositely Doppler-shifted responses), and the mechanism of sign reversal is therefore identical.

78.2 The averaged torque and the viscosity

Step 1. $\boldsymbol{M} \wedge \boldsymbol{H} = -M_y H_x$, so with (2D-11),

$$\boldsymbol{M} \wedge \boldsymbol{H} = -\tau_B \chi H_0^2\, \Omega \cos^2\alpha\, \cos\omega t \cos(\omega t - 2\alpha), \tag{78.2.1}$$

and averaging with $\overline{\cos^2\omega t} = \tfrac{1}{2}$, $\overline{\sin\omega t \cos\omega t} = 0$,

$$\overline{\boldsymbol{M} \wedge \boldsymbol{H}} = -\tfrac{1}{2}\tau_B \chi H_0^2\, \Omega \cos^2\alpha \cos 2\alpha. \tag{78.2.2}$$

Step 2: the prefactor. With the 2D values,

$$\tau_B \chi H_0^2 = \frac{4\eta V}{kT} \cdot \frac{nm^2}{2kT} \cdot H_0^2 = 2\eta\,(nV)\,\frac{m^2 H_0^2}{(kT)^2} = 2\eta\varphi\,\xi_0^2 = 4\eta\varphi\,\xi^2, \tag{78.2.3}$$

where $\xi_0 = mH_0/kT$ and, following the papers' Sec. II convention, $\xi = \xi_0/\sqrt{2}$ is the parameter built on the acting field $H_0/\sqrt{2}$, so that $\xi_0^2 = 2\xi^2$. (In three dimensions the same combination was $2\eta\varphi\xi^2$.)

Step 3: the slip and the viscosity. From (2D-7), $\Omega - \Theta = -\overline{\boldsymbol{M} \wedge \boldsymbol{H}}/(4\eta\varphi)$, so

$$g = \frac{\Omega - \Theta}{\Omega} = \frac{\frac{1}{2} \cdot 4\eta\varphi\xi^2 \cos^2\alpha \cos 2\alpha}{4\eta\varphi} = \tfrac{1}{2}\xi^2 \cos^2\alpha \cos 2\alpha, \tag{78.2.4}$$

and by (2D-8), $\Delta\eta = \eta\varphi g$:

$$\boxed{\Delta\eta = \tfrac{1}{2}\,\eta\varphi\,\xi^2 \cos^2\alpha\, \cos 2\alpha} \tag{2D-12}$$

Substituting $\cos^2\alpha = (1+\omega^2\tau_B^2)^{-1}$ and $\cos 2\alpha = (1-\omega^2\tau_B^2)/(1+\omega^2\tau_B^2)$:

$$\boxed{\Delta\eta = \frac{1}{2}\,\eta\varphi\,\xi^2\,\frac{1-\omega^2\tau_B^2}{\left(1+\omega^2\tau_B^2\right)^2}, \qquad \tau_B = \frac{4\eta V}{kT}} \tag{2D-13}$$

the two-dimensional counterpart of (SM-28).

78.3 Universality of the frequency structure

Compare (2D-13) with (SM-28): *the frequency-dependent factor is identical.* Hence, without any further calculation,

Feature	3D	2D	Status
sign change	$\omega_0\tau_B = 1$	$\omega_0\tau_B = 1$	identical
minimum at	$\omega_m\tau_B = \sqrt{3}$	$\omega_m\tau_B = \sqrt{3}$	identical
depth	$-\Delta\eta(0)/8$	$-\Delta\eta(0)/8$	identical
amplitude $\Delta\eta(0)$	$\frac{1}{4}\eta\varphi\xi^2$	$\frac{1}{2}\eta\varphi\xi^2$	factor 2
value of τ_B	$3\eta V/kT$	$4\eta V/kT$	factor 4/3

Result 78.3.1 (General d)**.**

$$\Delta\eta = \frac{\eta\varphi\,\xi^2}{2(d-1)}\,\frac{1-\omega^2\tau_B^2}{\left(1+\omega^2\tau_B^2\right)^2}, \qquad \tau_B = \frac{2d\,\eta V_d}{(d-1)\,kT}. \tag{78.3.1}$$

Dimensionality enters only through the prefactor and through the *value* of τ_B; the *shape* of $\Delta\eta(\omega\tau_B)$ is dimension-independent.

Why the shape is universal. The frequency dependence comes entirely from Debye relaxation of a single scalar mode — amplitude reduced by $(1+\omega^2\tau_B^2)^{-1/2}$, phase lagged by $\arctan\omega\tau_B$ — and from the fact that the transverse response is the frequency-derivative of that mode, doubling the phase. Neither statement refers to the dimension of space. What dimension controls is only *how long* τ_B is and *how strongly* the resulting torque couples to the shear.

78.4 The high-frequency asymptote

For $\omega\tau_B \gg 1$, (2D-13) gives $\Delta\eta \to -\tfrac{1}{2}\eta\varphi\xi^2/(\omega\tau_B)^2$. With $\xi^2 = m^2H_0^2/2(kT)^2$, $\tau_B = 4\eta V/kT$ and $m = M_sV$:

$$\frac{\xi^2}{\tau_B^2} = \frac{m^2H_0^2}{2(kT)^2}\cdot\frac{(kT)^2}{16\eta^2V^2} = \frac{M_s^2H_0^2}{32\,\eta^2}, \tag{78.4.1}$$

the particle area cancelling exactly as the volume did in three dimensions. Hence, with the acting field $H = H_0/\sqrt{2}$,

$$\boxed{\Delta\eta = -\frac{\varphi}{\eta}\,\frac{M_s^2H^2}{32\,\omega^2}} \qquad (\text{3D: } -\frac{\varphi}{\eta}\Big(\frac{M_sH}{6\omega}\Big)^2 = -\frac{\varphi M_s^2H^2}{36\eta\omega^2}). \tag{2D-14}$$

τ_B again drops out, for the same reason: the falling amplitude and the growing lag cancel. The 2D coefficient is 1/32 against 1/36 — only 12% larger, because the factor 2 in the prefactor is nearly offset by the $(4/3)^2$ in τ_B^2.

Chapter 79

Arbitrary Amplitude

79.1 The F–G system

Write $M_x = nmF(t)$ and $M_y = nm\,\Omega\tau_B\,G(t)$ with $\xi(t) = \xi\gamma(t)$, exactly as in Ch. 51. Taking components of (2D-2) with (2D-3) and discarding $O((\Omega\tau_B)^2)$:

Longitudinal. The convective and nonlinear contributions to the x equation are both second order, leaving

$$\boxed{\tau_B \frac{\mathrm{d}F}{\mathrm{d}t} = L_2\big[\xi\gamma(t)\big] - F(t)} \tag{2D-15}$$

identical in form to (SM-36) with the 3D Langevin function replaced by I_1/I_0.

Transverse. The y equation reads $\mathrm{d}M_y/\mathrm{d}t = \Theta M_x - M_y/\tau_B$ with $\Theta = \Omega - M_y H/(4\eta\varphi)$; substituting and dividing by $nm\Omega$,

$$\tau_B \frac{\mathrm{d}G}{\mathrm{d}t} = F - G - \left[\frac{nm\tau_B H}{4\eta\varphi}\right] FG. \tag{79.1.1}$$

The bracket is evaluated with the 2D values:

$$\frac{nm\tau_B H}{4\eta\varphi} = \frac{n\,m}{4\eta\,(nV)} \cdot \frac{4\eta V}{kT} \cdot H = \frac{mH}{kT} = \xi\gamma(t), \tag{79.1.2}$$

against $\frac{1}{2}\xi\gamma$ in three dimensions — the general-d factor $1/(d-1)$. Hence

$$\boxed{\tau_B \frac{\mathrm{d}G}{\mathrm{d}t} = F - \Big(1 + \xi\gamma(t)F(t)\Big)G} \tag{2D-16}$$

Internal consistency check. In a stationary field, $\gamma = 1$ and $F \to L_2(\xi)$, so the bracket becomes $1 + \xi L_2 = 1 + \zeta$ with ζ exactly as found independently in (77.1.4). ✓ The same check in three dimensions gave $1 + \frac{1}{2}\xi L_3 = 1 + \zeta_{3\mathrm{D}}$.

79.2 Torque, viscosity, and the general formula

Averaging $\boldsymbol{M} \wedge \boldsymbol{H} = -M_y H_x$ over a period,

$$\overline{\boldsymbol{M} \wedge \boldsymbol{H}} = -nm\,\Omega\,\tau_B\,H_0\,\overline{G\gamma} = -4\eta\varphi\,\xi_0\,\Omega\,\overline{G\gamma}, \tag{2D-17}$$

using $nm\tau_B H_0 = 4\eta\varphi\xi_0$ from the same cancellation as (79.1.2). Then from (2D-7) and (2D-8),

$$\frac{\Omega - \Theta}{\Omega} = \xi_0\,\overline{G\gamma}, \qquad \boxed{\Delta\eta = \eta\varphi\,\xi_0\,\overline{G(t)\gamma(t)}} \tag{2D-18}$$

against $\Delta\eta = \frac{3}{4}\eta\varphi\xi\overline{G\gamma}$ and $(\Omega-\Theta)/\Omega = \frac{1}{2}\xi\overline{G\gamma}$ in three dimensions. General d: $(\Omega-\Theta)/\Omega = \xi\overline{G\gamma}/(d-1)$ and $\Delta\eta = \frac{d}{2(d-1)}\eta\varphi\xi\overline{G\gamma}$. ✓

79.3 The square-wave solution

For a rectangular waveform and weak fields the nonlinear term of (2D-16) is dropped and $L_2 \to \xi\gamma/2$. The resulting pair is *structurally identical* to the 3D pair of Sec. 54.5, with the single replacement $\xi/3 \to \xi/2$ in the drive of F. Since that constant factors out of the entire linear calculation, the result of Sec. 54.5 may be quoted with the coefficient changed:

$$\overline{G\gamma} = \frac{\xi}{2}\left(2 - \tanh^2\varepsilon - \frac{2\tanh\varepsilon}{\varepsilon}\right), \qquad \varepsilon = \frac{\pi}{2\omega\tau_B}, \tag{79.3.1}$$

and therefore, by (2D-18),

$$\boxed{\Delta\eta = \frac{1}{2}\,\eta\varphi\,\xi^2\left(2 - \tanh^2\varepsilon - \frac{2\tanh\varepsilon}{\varepsilon}\right)} \tag{2D-19}$$

Checks. As $\varepsilon \to \infty$ the bracket $\to 1$ and $\Delta\eta \to \frac{1}{2}\eta\varphi\xi^2$, which is the static weak-field result (77.3.1) obtained independently in Ch. 77. ✓ As $\varepsilon \to 0$ the bracket $\to -\varepsilon^2/3 < 0$, reproducing the negative high-frequency tail. ✓ The three characteristic numbers are *unchanged* from three dimensions, since the bracket is the same function:

$$\omega_0\tau_B = 0.978, \qquad \omega_m\tau_B = 1.707, \qquad \frac{\Delta\eta_m}{\Delta\eta(0)} = -0.105. \tag{79.3.2}$$

(The 3D misprint corrected in Volume VI — $\tanh\varepsilon$ for $\tanh^2\varepsilon$ — applies here identically, and the corrected form is used.)

79.4 Can the total viscosity vanish in two dimensions?

The condition $\eta_{\text{total}} = \eta + \Delta\eta = 0$ means $\Delta\eta = -\eta$, i.e. $g = -1/\varphi$. Normalised to the saturation value (76.2.4),

$$\frac{\Delta\eta}{\Delta\eta(\infty,0)} = -\frac{1}{\varphi} \qquad (\text{3D: } -\frac{2}{3\varphi}). \tag{79.4.1}$$

At $\varphi = 0.2$ this is -5, against $-10/3$ in three dimensions. **The two-dimensional pump is harder, not easier**, by a factor 3/2 — because the saturation increment $\eta\varphi$ against which everything is normalised is itself smaller by that factor, while the viscosity to be cancelled is unchanged. This is worth stating because the opposite is often assumed: reduced dimensionality amplifies fluctuations but does *not*, here, amplify the mean response.

Part XIV

Results, Comparisons and Practice

Chapter 80

Complete Governing Equations for the Two-Dimensional System

This chapter collects, in self-contained form, everything needed to work with or simulate the two-dimensional theory. Indices run over $1, 2$; Ω, Θ, Λ and $\boldsymbol{M} \wedge \boldsymbol{H}$ are pseudoscalars; ε_{ij} is the two-index Levi-Civita symbol with $\varepsilon_{12} = +1$.

80.1 Definitions and material constants

$$V = \pi a^2, \qquad \varphi = nV, \qquad m = M_s V, \tag{80.1.1}$$

$$\zeta_r = 4\pi\eta a^2 = 4\eta V, \qquad n\zeta_r = 4\eta\varphi, \qquad D_r = \frac{kT}{\zeta_r}, \tag{80.1.2}$$

$$\tau_B = \frac{1}{D_r} = \frac{4\eta V}{kT}, \qquad \tau_s = \frac{\rho_s d^2}{32\eta}, \qquad \xi = \frac{mH}{kT}, \tag{80.1.3}$$

$$L_2(\xi) = \frac{I_1(\xi)}{I_0(\xi)}, \qquad \chi = \frac{nm^2}{2kT}, \qquad \boldsymbol{M}_0 = nmL_2(\xi)\frac{\boldsymbol{H}}{H}. \tag{80.1.4}$$

80.2 Kinematics and constraints

$$\Omega = \tfrac{1}{2}\varepsilon_{ij}\partial_i v_j, \qquad \nabla\cdot\boldsymbol{v} = 0, \tag{80.2.1}$$

$$\varepsilon_{ij}\partial_i H_j = 0, \qquad \nabla\cdot\big(\boldsymbol{H} + 2\pi\boldsymbol{M}\big) = 0. \tag{80.2.2}$$

80.3 Momentum

$$\rho\left[\frac{\partial\boldsymbol{v}}{\partial t} + \big(\boldsymbol{v}\cdot\nabla\big)\boldsymbol{v}\right] = -\nabla p + \big(\boldsymbol{M}\cdot\nabla\big)\boldsymbol{H} + \eta\nabla^2\boldsymbol{v} + \tfrac{1}{2}\varepsilon_{ij}\partial_j\big(\boldsymbol{M}\wedge\boldsymbol{H}\big)\hat{\boldsymbol{x}}_i, \tag{80.3.1}$$

where the last term is written componentwise as $f_i = \tfrac{1}{2}\varepsilon_{ij}\partial_j(\boldsymbol{M}\wedge\boldsymbol{H})$.

80.4 Spin and magnetisation

$$I\frac{\mathrm{d}\Theta}{\mathrm{d}t} = \boldsymbol{M}\wedge\boldsymbol{H} - \frac{I}{\tau_s}\big(\Theta - \Omega\big) \;\xrightarrow[\tau_s\to 0]{}\; \Theta = \Omega + \frac{\boldsymbol{M}\wedge\boldsymbol{H}}{4\eta\varphi}, \tag{80.4.1}$$

$$\frac{\mathrm{d}M_i}{\mathrm{d}t} = \Theta\,\varepsilon_{ji}M_j - \frac{1}{\tau_B}\big(M_i - M_{0i}\big) \qquad \text{(phenomenological closure)}, \tag{80.4.2}$$

or, with the effective-field closure and $\boldsymbol{n} = \boldsymbol{\xi}_e/\xi_e$, $L_e = L_2(\xi_e)$,

$$\frac{\mathrm{d}}{\mathrm{d}t}\big(L_e n_i\big) = \Omega\,\varepsilon_{ji}L_e n_j - \frac{1}{\tau_B}\frac{L_e}{\xi_e}\big(\xi_{e,i} - \xi_i\big) - \frac{1}{\tau_B}\left(1 - \frac{2L_e}{\xi_e}\right)\Big[n_i\big(\boldsymbol{n}\cdot\boldsymbol{\xi}\big) - \xi_i\Big]. \tag{80.4.3}$$

80.5 Viscosity

$$\boxed{\Delta\eta = \eta\varphi\,\frac{\Omega - \Theta}{\Omega}, \qquad \overline{\boldsymbol{M}\wedge\boldsymbol{H}} = -4\eta\varphi\big(\Omega - \Theta\big)} \tag{80.5.1}$$

Stationary field.

$$\Delta\eta = \eta\varphi\,\frac{\xi L_2(\xi)}{1 + \xi L_2(\xi)} \;\longrightarrow\; \begin{cases} \tfrac{1}{2}\eta\varphi\xi^2, & \xi \ll 1, \\ \eta\varphi, & \xi \gg 1. \end{cases} \tag{80.5.2}$$

Alternating field $\boldsymbol{H} = (H_0\cos\omega t, 0)$, weak amplitude.

$$\Delta\eta = \frac{1}{2}\eta\varphi\xi^2\,\frac{1 - \omega^2\tau_B^2}{\big(1 + \omega^2\tau_B^2\big)^2}, \qquad \xi = \frac{mH_0}{\sqrt{2}\,kT}, \tag{80.5.3}$$

with zero at $\omega\tau_B = 1$, minimum $-\Delta\eta(0)/8$ at $\omega\tau_B = \sqrt{3}$, and the high-frequency tail

$$\Delta\eta \to -\frac{\varphi M_s^2 H^2}{32\,\eta\,\omega^2}. \tag{80.5.4}$$

Alternating field, arbitrary amplitude. Solve

$$\tau_B\dot{F} = L_2\big[\xi\gamma(t)\big] - F, \qquad \tau_B\dot{G} = F - \big(1 + \xi\gamma(t)F\big)G, \tag{80.5.5}$$

for the periodic state, then

$$\Delta\eta = \eta\varphi\,\xi\,\overline{G\gamma}, \qquad \overline{G\gamma} = \frac{\omega}{2\pi}\int_0^{2\pi/\omega} G\gamma\,\mathrm{d}t = \frac{\omega}{\pi}\int_0^{\pi/\omega} G\,\mathrm{d}t \;\; \text{(rectangular } \gamma). \tag{80.5.6}$$

80.6 Master consistency check: general d

Every boxed result above is the $d = 2$ member of a family whose $d = 3$ member is the published theory:

$$\zeta_r = 2d\,\eta V_d, \qquad \tau_B = \frac{2d\,\eta V_d}{(d-1)kT}, \qquad \chi = \frac{nm^2}{d\,kT}, \tag{80.6.1}$$

$$\Delta\eta = \frac{d}{2}\eta\varphi\,\frac{\Omega - \Theta}{\Omega}, \qquad \zeta = \frac{\xi L_d}{d-1}, \qquad \Delta\eta\big|_{\xi\ll1} = \frac{\eta\varphi\xi^2}{2(d-1)}. \tag{80.6.2}$$

Substituting $d = 3$ reproduces (SM-5), (SM-15), (SM-34), (SM-2) and (SM-3) respectively. ✓

Chapter 81

Summary of Every Modification from 3D to 2D

Quantity	3D	2D	Reason for the change
ε identity	$\varepsilon_{ijk}\varepsilon_{klm} = \delta\delta - \delta\delta$	$\varepsilon_{ij}\varepsilon_{kl} = \delta\delta - \delta\delta$; $\varepsilon_{ij}\varepsilon_{kj} = \delta_{ik}$	two indices, not three
cross product	vector	pseudoscalar $\boldsymbol{a} \wedge \boldsymbol{b}$	no normal direction in the plane
Ω, Θ	vectors	pseudoscalars	$SO(2)$ is one-dimensional
$\Omega = \frac{1}{2}\operatorname{curl} \boldsymbol{v}$	vector identity	same formula, scalar output	unchanged algebra, changed type
$\nabla\times\nabla\times$	$\nabla(\nabla\cdot) - \nabla^2$	$\varepsilon_{ij}\partial_j\Omega = -\frac{1}{2}\nabla^2 v_i$	curl changes type
isotropic $\langle nn\rangle$	$\frac{1}{3}\delta_{ij}$	$\frac{1}{2}\delta_{ij}$	$\delta_{ii} = 2$
Green's function	$-1/4\pi r$	$\ln r/2\pi$	radial solution of $\nabla^2 G = 0$
translational drag	$6\pi\eta a$	**does not exist**	Stokes paradox
ζ_r	$8\pi\eta a^3 = 6\eta V$	$4\pi\eta a^2 = 4\eta V$	rotlet flow, $\sigma \propto r^{-2}$
I_p	$\frac{2}{5}m_p a^2$	$\frac{1}{2}m_p a^2$	disk, not sphere
τ_s	$\rho_s d^2/60\eta$	$\rho_s d^2/32\eta$	new I_p and ζ_r
orientation measure	$\sin\theta\,\mathrm{d}\theta\,\mathrm{d}\phi$	$\mathrm{d}\phi$	circle has no latitudes
partition function	$4\pi\sinh\xi/\xi$	$2\pi I_0(\xi)$	no $u = \cos\theta$ substitution
Langevin function	$\coth\xi - 1/\xi$	$I_1(\xi)/I_0(\xi)$	new measure
weak-field L	$\xi/3$	$\xi/2$	$\langle\cos^2\rangle = 1/d$
strong-field L	$1 - 1/\xi$	$1 - 1/2\xi$	fewer transverse modes
$\langle\cos^2\rangle$	$1 - 2L/\xi$	$1 - L/\xi$	Bessel recurrence
$\langle e_i e_j\rangle$	$\frac{L}{\xi}\delta + (1 - \frac{3L}{\xi})nn$	$\frac{L}{\xi}\delta + (1 - \frac{2L}{\xi})nn$	trace = 2, not 3
χ	$nm^2/3kT$	$nm^2/2kT$	weak-field L
$\nabla_s^2 \boldsymbol{e}$	$-2\boldsymbol{e}$	$-\boldsymbol{e}$	eigenvalue $-(d-1)$
τ_B	$3\eta V/kT$	$4\eta V/kT$	new ζ_r *and* new eigenvalue
$D_r = kT/\zeta_r$	—	unchanged	Einstein argument is dimensionless
moment equation	$\frac{1}{2\tau_B}[\xi - \langle ee\rangle\xi]$	$\frac{1}{\tau_B}[\cdots]$	both equal D_r
$n\zeta_r$	$6\eta\varphi$	$4\eta\varphi$	new ζ_r

Quantity	3D	2D	Reason for the change
couple → force	$\frac{1}{2}\nabla\times\mathbf{\Lambda}$	$\frac{1}{2}\varepsilon_{ij}\partial_j\Lambda$	pseudoscalar couple
master formula	$\frac{3}{2}\eta\varphi g$	$\eta\varphi g$	factor $d/2$
ζ (stationary)	$\frac{1}{2}\xi L_3$	ξL_2	factor $1/(d-1)$
$\Delta\eta$ static	$\frac{3}{2}\eta\varphi\frac{\xi-\tanh\xi}{\xi+\tanh\xi}\sin^2\beta$	$\eta\varphi\frac{\xi L_2}{1+\xi L_2}$	no elementary form; no β
weak-field $\Delta\eta$	$\frac{1}{4}\eta\varphi\xi^2$	$\frac{1}{2}\eta\varphi\xi^2$	$1/2(d-1)$
saturation	$\frac{3}{2}\eta\varphi$	$\eta\varphi$	factor $d/2$
$\sin^2\beta$ factor	present	**absent**	$\boldsymbol{H}\perp\Omega$ always
alternating $\Delta\eta$	$\frac{1}{4}\eta\varphi\xi^2\mathcal{F}$	$\frac{1}{2}\eta\varphi\xi^2\mathcal{F}$	prefactor only
$\mathcal{F}(\omega\tau_B)$	$\frac{1-\omega^2\tau_B^2}{(1+\omega^2\tau_B^2)^2}$	identical	Debye response is scalar
$\omega_0\tau_B$, $\omega_m\tau_B$	1, $\sqrt{3}$	1, $\sqrt{3}$	unchanged
HF asymptote	$-\varphi M_s^2H^2/36\eta\omega^2$	$-\varphi M_s^2H^2/32\eta\omega^2$	new τ_B, prefactor
G-equation term	$(1+\frac{1}{2}\xi\gamma F)G$	$(1+\xi\gamma F)G$	factor $1/(d-1)$
$\Delta\eta$ from $\overline{G\gamma}$	$\frac{3}{4}\eta\varphi\xi\overline{G\gamma}$	$\eta\varphi\xi\overline{G\gamma}$	$\frac{d}{2(d-1)}$
square wave	$\frac{1}{4}\eta\varphi\xi^2 f(\varepsilon)$	$\frac{1}{2}\eta\varphi\xi^2 f(\varepsilon)$	prefactor only
pump condition	isoline $-2/3\varphi$	isoline $-1/\varphi$	smaller saturation
B–H relation	$B=H+4\pi M$	$B=H+2\pi M$	S_d: $4\pi\to2\pi$
dipole field	$\sim m/r^3$	$\sim m/r^2$	2D Green's function

Chapter 82

Mathematical Differences Between 3D and 2D

82.1 Algebra

The three-index Levi-Civita symbol is replaced by a two-index one, and with it every contraction identity. The cross product ceases to be a binary operation on vectors: it becomes a pairing of two vectors into a pseudoscalar, together with a separate pairing of a pseudoscalar and a vector into a vector. Nested products must therefore be written with care, and the naive transcription of $\boldsymbol{a}\times(\boldsymbol{a}\times\boldsymbol{b})$ reverses sign (Sec. 71.3). Antisymmetric second-rank tensors have one independent component instead of three, which is why the spin field is a scalar.

82.2 Calculus

The curl changes type and is no longer an endomorphism; the identity $\nabla\times\nabla\times=\nabla\nabla\cdot-\nabla^2$ is replaced by (71.4.1). The Laplacian, gradient and divergence are unchanged in form. On the orientation manifold the change is a simplification: the surface operators of the sphere collapse to ∂_ϕ, with no metric factor, and Green's identity becomes integration by parts over a period.

82.3 Measures, transforms and special functions

The orientation measure loses its $\sin\theta$; the total solid angle is 2π instead of 4π; Fourier integrals carry $\mathrm{d}^2k/(2\pi)^2$; the delta function has units cm^{-2}. The Laplace Green's function becomes logarithmic, which is the root of the Stokes paradox and of the long-time-tail problem. The equilibrium orientational integrals cease to be elementary and become modified Bessel functions — the single most visible change in the final formulae. The useful identities are

$$\int_0^{2\pi} e^{\xi\cos\phi}\mathrm{d}\phi=2\pi I_0,\quad I_0'=I_1,\quad I_0-I_2=\frac{2}{\xi}I_1,\quad I_\nu\sim\frac{e^\xi}{\sqrt{2\pi\xi}}\Big(1-\frac{4\nu^2-1}{8\xi}\Big).\tag{82.3.1}$$

82.4 Eigenvalues and degeneracies

The vector harmonics on S^{d-1} have Laplacian eigenvalue $-(d-1)$ and degeneracy d: on the sphere, -2 with degeneracy 3; on the circle, -1 with degeneracy 2. The relaxation rate of the first moment is $(d-1)D_r$, and this is where the factor of two in τ_B is lost. It is the change most

likely to be missed, because nothing in the appearance of the formula $\tau_B = 3\eta V/kT$ advertises that the 3 contains a factor of $\frac{1}{2}$ from the eigenvalue and a factor 6 from the friction.

82.5 Where a naive reduction would fail

1. $\varepsilon_{ijk}\varepsilon_{ljk} = 2\delta_{il}$ becomes $\varepsilon_{ij}\varepsilon_{kj} = \delta_{ik}$, not $2\delta_{ik}$.
2. τ_B would be wrong by $4/3$ if either the friction or the eigenvalue were carried over.
3. $L_3 \to L_2$ is not a change of coefficient but a change of function class: hyperbolic to Bessel. Any formula relying on the identity $L = (\xi - \tanh\xi)/(\xi\tanh\xi)$ — in particular the compact form of (SM-2) — has no 2D counterpart.
4. The coefficient $1 - 3L/\xi$ in the effective-field equation becomes $1 - 2L/\xi$; taking it as $1 - 3L_2/\xi$ violates the trace of $\langle e_i e_j\rangle$.
5. $\sin^2\beta$ must be *deleted*, not set to a value: the parameter β does not exist.
6. The nested-wedge transcription of the transverse relaxation term reverses its sign.
7. $6\pi\eta a$ has no counterpart at all.

Chapter 83

Physical Differences Between 3D and 2D

83.1 What is unchanged, and why that is the main result

The *mechanism* is untouched. A linearly polarised field still picks out an axis but no sense of rotation; a shear still breaks that degeneracy by Doppler-shifting the two circular components to $\omega \mp \Omega$; the transverse magnetisation still lags by 2α; the averaged torque still reverses sign when α passes $\pi/4$. Consequently the frequency at which the viscosity changes sign, the frequency of the deepest negative excursion, and the relative depth of that excursion are *identical* in two and three dimensions. Dimensionality rescales the effect; it does not reshape it.

83.2 What changes quantitatively

At equal ξ and equal area/volume fraction, the weak-field rotational viscosity is twice as large in two dimensions, while the saturation value is smaller by $\frac{3}{2}$. The Brownian time is longer by $4/3$ at equal ηV. The approach of the magnetisation to saturation is twice as fast in $1/\xi$.

83.3 What changes structurally

- **The angular degree of freedom disappears.** With $\boldsymbol{H}$ necessarily perpendicular to Ω, the response depends on $(\xi, \omega\tau_B)$ alone. The 2D theory is a genuinely two-parameter theory; the 3D one is three-parameter.

- **Translational hydrodynamics is ill posed.** The Stokes paradox means that the mobility of a disk is not defined in an unbounded 2D fluid. The magnetoviscous theory survives because it never uses it — but any extension to sedimentation, field-gradient-driven transport (the Kelvin force term), or diffusiophoresis *does* require a cutoff, supplied physically by the film thickness or the container.

- **The viscosity itself is only conditionally defined.** See Sec. 83.4.

83.4 Fluctuations: long-time tails and the logarithmic viscosity

In two dimensions the velocity autocorrelation function of a tagged particle decays as t^{-1} rather than $t^{-3/2}$, and the Green–Kubo integral for the shear viscosity diverges logarithmically with system size. Strictly speaking, therefore, a two-dimensional fluid does not possess a finite shear viscosity in the thermodynamic limit: η acquires a $\ln(L/a)$ dependence on the system size L.

What this means for this volume. Every formula here is written in terms of η, and is correct provided η is understood as the viscosity measured on the scale of the experiment. Two remarks make this precise:

- The divergence is logarithmic, hence weak: a change of system size by $e \approx 2.7$ changes η by one increment of the (small) coefficient.
- In the physically realisable quasi-2D case, the surrounding bulk fluid cuts the divergence off at the Saffman–Delbrück length (72.7.1), and η is finite and well defined. This is the strongest argument for regarding the thin film, rather than the strictly 2D world, as the physical realisation.

Note that $\Delta\eta/\eta$, which is what the theory actually predicts, is a *ratio* and is far less sensitive to this ambiguity than $\Delta\eta$ itself.

83.5 Order, symmetry and the Mermin–Wagner theorem

In two dimensions a system with short-range interactions and a continuous symmetry cannot develop true long-range order at finite temperature. Three comments:

1. **It does not affect the theory as formulated.** The moments here are non-interacting; the magnetisation is induced by an external field, not spontaneous. An applied field breaks the symmetry explicitly, and the Mermin–Wagner theorem says nothing about explicitly broken symmetry.
2. **It would affect an interacting extension.** A 2D ferrofluid with strong dipolar coupling could not order ferromagnetically in the way a 3D one can — except that dipolar interactions are long-ranged, and the theorem's hypothesis fails. The correct statement is that the question is open at the level of this theory and requires the interacting problem.
3. **Positional order is genuinely different.** A 2D colloid has only quasi-long-range positional order and melts by the KTHNY route through a hexatic phase. At $\varphi = 0.2$ this is irrelevant; at high area fraction it would replace the "concentrated suspension" corrections of Sec. 15.6 with something qualitatively different.

83.6 Interaction range

The dipolar field decays as r^{-2} in two dimensions against r^{-3} in three — apparently longer-ranged. But the number of neighbours grows as r^{d-1}, and the two effects cancel: the mean dipolar energy is logarithmically sensitive to the cutoffs in *both* dimensions. The dilute approximation is therefore no more suspect in 2D than in 3D, contrary to a natural first guess.

83.7 Conserved quantities and topology

Mass, linear momentum and energy are conserved as before. Angular momentum is now a single scalar, and the exchange between spin and orbital angular momentum — the heart of the effect — is a scalar exchange. One genuinely new conserved structure appears in 2D ideal flow (the infinite family of enstrophy-like invariants), but it is destroyed by viscosity and by the spin coupling, and plays no role here. The topology of the orientation manifold changes from S^2 (simply connected) to S^1 (not simply connected): winding number is now a meaningful label for a trajectory of the moment, which matters for the rotating-field problem, where a particle in a strong circularly polarised field executes genuine 2π rotations.

Chapter 84

Implementation Notes

84.1 Operators and discretisation

- Store Ω and Θ as *scalar* fields. Attempting to keep three components wastes memory and invites sign errors.

- Implement the curl once, in both directions: $\operatorname{curl}\boldsymbol{v} = \partial_x v_y - \partial_y v_x$ (vector $\to$ scalar) and $(\operatorname{curl} S)_i = \varepsilon_{ji}\partial_j S$ (scalar $\to$ vector). Use (71.4.1) as a unit test on the discrete operators: the identity $\varepsilon_{ij}\partial_j\Omega = -\tfrac{1}{2}\nabla^2 v_i$ should hold to round-off for a divergence-free discrete field, and it will not hold if the two curls use inconsistent staggering.

- The stream-function–vorticity formulation is particularly natural here, since Ω is already the primary scalar: $\nabla^2\psi = -2\Omega$, $\boldsymbol{v} = (\partial_y\psi, -\partial_x\psi)$. The magnetic coupling enters as a source in the vorticity equation.

84.2 Special functions

$L_2 = I_1/I_0$ should not be evaluated by calling two Bessel routines and dividing for large ξ: both overflow like e^{ξ}. Use the exponentially scaled forms $\tilde{I}_\nu = e^{-\xi}I_\nu$, which most libraries provide, or the continued fraction for the ratio. For $\xi < 10^{-4}$ use the series $\xi/2 - \xi^3/16$ to avoid cancellation; for $\xi > 50$ use $1 - 1/2\xi - 1/8\xi^2$. A useful check is $0 < L_2(\xi) < \min(\xi/2, 1)$ for all $\xi > 0$.

Similarly, $(\mathrm{d}\ln L_2/\mathrm{d}\xi)^{-1}$ in (80.4.3) is best evaluated from

$$\frac{\mathrm{d}L_2}{\mathrm{d}\xi} = 1 - L_2^2 - \frac{L_2}{\xi}, \tag{84.2.1}$$

which follows from the Bessel recurrences and avoids numerical differentiation. (Check: as $\xi \to 0$ this gives $\mathrm{d}L_2/\mathrm{d}\xi \to 1 - 0 - 1/2 = 1/2$ ✓; as $\xi \to \infty$, $L_2 \to 1 - 1/2\xi$ gives $1 - (1 - 1/\xi) - 1/\xi + \cdots = O(\xi^{-2})$ ✓.)

84.3 Integrating the F–G system

- The system (80.5.5) is stiff when $\xi \gg 1$, because the effective relaxation rate of G is $(1 + \xi\gamma F)/\tau_B$, which at $\xi = 50$ is fifty times $1/\tau_B$. Use an implicit or exponential integrator, or a step $\Delta t \lesssim \tau_B/(10\,\xi)$ with explicit RK4.

- Discard several periods before averaging: the initial condition $F = G = 0$ is not on the periodic orbit, and the transient decays at the same stiff rate.

- Exploit antiperiodicity, $F(t + T/2) = -F(t)$ and $G(t + T/2) = -G(t)$, both as a convergence diagnostic and to halve the averaging integral.

- For a rectangular waveform, do not let a time step straddle a switching instant: place grid points exactly on the discontinuities, or the effective γ is smeared and the measured $\omega_0\tau_B$ drifts.

84.4 Normalisations and units in code

Work in the dimensionless variables ξ, $\omega\tau_B$, φ and $\Delta\eta/\eta$; then no factor of 2π from the magnetic convention, and no choice of η units, ever enters. The only place dimensional quantities are needed is in converting an experimental frequency to $\omega\tau_B$, for which $\tau_B = 4\eta V/kT$ — *not* $3\eta V/kT$ — must be used. Using the 3D formula would misplace the sign-change frequency by 33%.

84.5 Boundary conditions

- Velocity: no-slip on solid boundaries, as usual.

- Spin: because Θ is determined algebraically by (80.4.1) in the inertialess limit, it requires *no* boundary condition. Only if the inertial term is retained (which Sec. 72.5 shows is never necessary) would a spin boundary condition be needed.

- Magnetisation: (80.4.2) is first order in time and contains no spatial derivatives of $\boldsymbol{M}$, so it needs an initial condition but no boundary condition. This is a real simplification over models with magnetisation diffusion.

- Magnetic field: if the applied field is taken as uniform — the case treated throughout — (80.2.2) is satisfied identically and no magnetostatic solve is required. A non-uniform field requires solving $\nabla^2\psi = 2\pi\nabla\cdot\boldsymbol{M}$ with the 2D Green's function, where the logarithmic kernel makes the far field sensitive to the domain size; use periodic boundaries and Ewald-type summation, or subtract the mean magnetisation.

84.6 Stability and diagnostics

- The effective viscosity $\eta(1 + \varphi g)$ appears in the momentum equation. If a parameter regime drives $\varphi g < -1$, the diffusion coefficient becomes negative and any explicit scheme will blow up — *as it physically should*, since that is precisely the self-pumping instability of Sec. 79.4. Distinguish numerical blow-up from physical instability by checking whether $\varphi g < -1$; by (79.4.1) this requires $|g| > 1/\varphi$, which the theory says is not reachable for realistic parameters.

- Useful invariants to monitor: $|F| \le 1$ always (it is a normalised magnetisation); $\Delta\eta \to \frac{1}{2}\eta\varphi\xi^2$ as $\omega \to 0$; $\Delta\eta$ must change sign at $\omega\tau_B = 1$ in the weak-field limit regardless of waveform up to 2%.

- A good end-to-end test of an implementation: set $\gamma = 1$ (static field) and verify $\Delta\eta = \eta\varphi\xi L_2/(1+\xi L_2)$ from the time-stepped solution, to the tolerance of the integrator.

End of Volume VIII. The two-dimensional theory is complete: every equation of Volumes I–VII has been reconsidered, and those that depend on dimensionality have been re-derived rather than transcribed.